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Mitigating the Financial Risk Behind Emission Cap Compliance: A Case in Maritime Transportation

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The enforcement of ever-stringent regulatory requirements capping the emission limits is challenging the traditional operations of various transportation sectors. In the case of maritime transportation, the recent regulation tightening the fuel sulphur limits to 0.50%, known as the "IMO 2020", has been enforced. There is a flurry of recent activities by ocean carriers to equip their vessels to comply with this new regulation. While the technical conditions of the compliance solutions are clear, decisions on which to adopt are hard to make due to the inevitable uncertainties in the current transition period, especially on the impact of the prices of different types of fuel in the long run. In this study, we consider the technology investment decisions towards emission cap compliance for an ocean carrier. Each compliance solution is subject to uncertain transportation demand and uncertain operating costs with a partially characterized probability distribution that may deviate from current expected norms. The ocean carrier chooses a portfolio of compliance solutions for its entire fleets that would best adhere to two sets of decision criteria characterized by (i) a net present value (NPV) target in the ship fleet upgrading investment and (ii) a capacity utilization rate (CUR) target in the ship fleet deployment. To find optimal decisions that will perform well in the uncertain transition period, we introduce a tractable mathematical model, termed the ambiguous robustness optimization (ARnO) model, to minimize the financial riskiness index (FRI) associated with the risk of expected NPV not meeting a specified target. We further propose a solution scheme through a mixed-integer second-order cone programming approximation that can be efficiently solved by off-the-shelf solvers. We show that this decision support system performs well in numerical experiments constructed using real data on the Asia-North America West Coast shipping network.

Key words: emission cap, sustainable operations, shipping finance, risk and ambiguity, robustness optimization, conic programming

1 Introduction

Emissions from ship combustion systems contributed significantly to global sulphur inventories, representing 13% of global anthropogenic Sulphur oxides (SOx) emissions from 2007 to 2012 (Smith et al. 2014). SOx emissions form sulphate aerosols that risk human health, acidify terrestrial and aquatic environments, and cause climate change and global warming[1]. The International Maritime Organization (IMO)[2] rolled out a landmark set of regulations, known as the *IMO 2020*, which substantially tighten the global cap of the sulphur content in ships' fuel oil from 3.50% m/m (mass by mass) to 0.50% m/m. It is expected to cost carriers at least 15 billion USD per year and bring far-reaching consequences throughout the environment and the global economy (Forbes 2019).

The IMO 2020 has wide-ranging repercussions in the shipping industry. Massive changes posed by the IMO 2020 are apparent throughout the shipping industry—shipowners must decide on which *compliance solution* to implement in their vessels to retain the seaworthiness of their vessels. Three types of compliance technologies are identified and widely recognized in the shipping industry (IHS Markit 2019, Forbes 2019, IMO 2020, Deloitte 2019), as shown in Figure EC.1: (1) Switching from high-sulphur fuel oil (HSFO) to cleaner but pricier marine fuels with a maximum sulphur content of 0.5%, such as very-low-sulphur fuel oil (VLSFO) or marine gas oil (MGO).(2) Installing exhaust gas cleaning systems (EGCS), also known as "scrubbers". These scrubbers are designed to remove sulphur oxides from the ship's engine and boiler exhaust gases. A ship fitted with a scrubber can still use HSFO, since it can reduce the sulphur oxides emissions to the level required by the fuel oil sulphur limit. (3) Retrofitting the vessel to use liquefied natural gas (LNG) containing almost zero sulphur. However, switching to LNG requires staff training and installing a different type of ship engine and special storage onboard, which is too costly, complicated, and not viable in the short term (Deloitte 2019).

Both scrubber installation and LNG retrofitting have an extended lead time. The former takes $24-36$ weeks (Deloitte 2019), while the latter takes around 60 weeks (Wärtsilä 2012). During this period, the vessel cannot be deployed. While the technological solutions to comply with the IMO 2020 are available, which technology to invest in is a hard decision for shipping companies. Although the IMO 2020 sulphur regulation has been enforced, among 60,000 vessels in operation today that fall into the compliance designation[3], only 4,000 (both operational and on order) vessels have been equipped with scrubbers by March 24, 2020, as estimated by industry sources[4]. In the

[1] See Hassellov IM, Turner DR, Lauer A, Corbett JJ, 2013 Shipping contributes to ocean acidification. Geophysical Research Letters 40(11):2731–2736.

[2] IMO is the United Nations specialized agency with responsibility for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships.

[3] Source:`https://www.hellenicshippingnews.com/lsfo-demand-to-displace-hsfo-in-asia-ahead-of-imo-2020/`, accessed July 9, 2020

[4] Source:`https://www.manifoldtimes.com/news/dnv-gl-estimates-4000-vessels-to-be-equipped-with-scrubbers/`, accessed July 9, 2020

current transition period, fuel price and quality will be the main factors of consideration for shipping companies (Deloitte (2019)). However, in the long run, investing in the right IMO 2020-compliant solutions to be future-ready is critical too. Figure 1 shows the sharp switch to low sulphur fuel oils (LSFO) in major bunkering hubs like Singapore and Panama in Q1 2020, although the volume of HSFO picks up slightly in Q1 2021[5]. "After the quick adjustment to the global sulphur cap, the decline in HSFO's share of total bunker sales has stopped... the steadily rise in scrubber-fitted ships has supported demand for HSFO and will remain until new solutions and future fuels are widely introduced on the industry's path to decarbonisation," says Peter Sand, BIMCO's Chief Shipping Analyst. In a nutshell, the industry is going through a transition period before the impact of IMO 2020 will be fully felt.

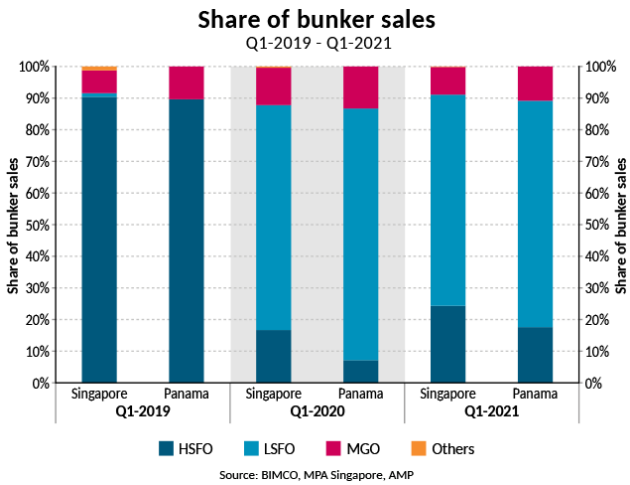


Figure 1 Share of bunker fuels in major bunkering hubs

Accompanied with their investments in technological compliant solutions, shipping companies are confronted with the elevated risk associated with uncertain bunker prices at the volatile global energy market, as bunker expenses constitute a high percentage of financial exposures after the initial investment. Figure EC.2 shows the high volatility of MGO, VLSFO and heavy fuel oil (HFO) prices of the global 20 ports average[6] over time.

The volatility of unexpected outcomes associated with investments in technological compliance solutions motivates the need for careful financial risk management. Still, a vessel lasts for about $20-30$ years before being scrapped in the demolition market (Stopford 2008). Consequently, the

[5] `https://www.bimco.org/news/market_analysis/2021/20210429-high-sulphur-fuel-oil-sales-rebound-after-pre-imo-2020-correction`, accessed 16 May 2021

[6] Global 20 Ports Average is, for each grade, the linear average of the following 20 major global bunkering ports, that together are responsible for a vast majority of global volumes: Busan, Colombo, Durban, Fujairah, Gibraltar, Hong Kong, Houston, Istanbul, LA/Long Beach, Las Palmas, Mumbai, New York, Panama, Piraeus, Rotterdam, Santos, Shanghai, Singapore, St Petersburg, Tokyo.

planning horizon for upgrading investments spans years or even decades, during which the energy markets are too volatile to be characterized accurately.

The deliberation on technology investment must also consider the demand side risk, since new technology is only helpful if suitably used. In liner shipping, an *ocean carrier* provides scheduled services with published itineraries to transport containerized goods by sea on a pre-determined service network through a fleet of container ships. The overarching challenge confronted by an ocean carrier is to make technology investment decisions to comply with the stringent emission cap while achieving two possibly conflicting objectives: (i) to maximize the net present value (NPV) of their investment in a volatile energy market and (ii) to maximize the utilization of the assets deployed. These decisions must be adopted to avoid interruptions to the existing services in ship fleet deployment.

This approach to the technology investment decisions is very complicated as the NPV is predicated on the service usage, which may change considerably with the new technology adoption. Fortunately, there is evidence that the adoption of compliance options in IMO 2020 will not affect the deployment plan for the vessels too much. Hence, existing scenarios can be used to construct a good proxy of the NPV and can be used to calibrate the impact of the decisions. A case in point is the Shipping Line Evergreen who will have around 140 vessels fitted with scrubbers among its 200-vessel fleet after the retrofitting and new-building scrubber program. No notable change in service routes has happened despite the enforcement of IMO 2020 since January 1, 2020. Instead, the company announced a low sulphur fuel compliance charge that gets customers to pay up the additional costs towards smooth operations[7]. Based on this assumption, we take the operating parameters (routing and sailing speed, etc.) as given according to the *status quo* and focus on the financial returns (savings in fuel costs) of the technology investment decisions.

An approach in practice: calculating payback time

Many consulting institutions and research institutes have published advisory notes or white papers to facilitate regulation-compliant decision-making (e.g., IHS Markit 2019, Deloitte 2019). Existing procedures are guided by scenario-based analysis for the calculation of payback time (e.g., Zis et al. 2016, IHS Markit 2019). Assumptions such as charter rates, fuel price differences, and vessel utilization rates are made to estimate the time to recover the cost of investment towards compliance. Although such analysis for payback time is a straightforward tool for investment decisions, it is not suitable for liner ship fleet upgrading for the following reasons:

[7] Source:`https://www.evergreen-line.com/tuf1/jsp/TUF1_News.jsp?newsType=G2&newsId=NEWS20190111000025 38&lang=en`, accessed July 9, 2020

- First, a fleet of vessels are managed jointly by an ocean carrier, but vessels are considered separately in the calculation of payback time.
- Second, the energy market is volatile. It is hard to forecast and assume bunker fuel prices accurately. Thus, a practical viable model should account for these uncertainties and provide a robust solution for decision-making.
- Third, the calculation of payback time failed to quantify the impact of retrofitting lead time and the waste of ship slots in twenty-foot equivalent units (TEUs).
- Finally, the calculation of payback time only provides an answer towards "how" but not "when". Existing advisory notes compare and analyze the different alternatives and aim to answer "how to comply with the IMO 2020". However, in liner ship fleet upgrading problems, time matters. For example, liner shipping companies may wish to send the vessel for retrofitting when the transportation demand is low to avoid unnecessary disruption to the operations, since such retrofitting takes time. For another example, the liner shipping companies will not send too many ships for retrofitting in one period, affecting the fleet capacity and profitability.

Decision-making under emission regulations is an emerging issue in maritime transportation (see, e.g., Balland et al. 2012, Fagerholt et al. 2015, Wang et al. 2020 and Wang et al. 2021). For example, Balland et al. (2013) considered uncertain emission reduction effects of emission controls and developed a two-stage stochastic program to minimize the cost of the emission control over a finite planning horizon. Our paper tackles this emerging issue from a technology investment perspective. We consider an ocean carrier that operates a fleet of ships and pays for the IMO 2020-compliant technology investment. We refer to this adaption problem of the existing fleet of liner ships in response to regulatory changes as *liner ship fleet upgrading (LSFU) problem.*

Our contributions

We summarize the contributions of this paper as follows:

- We scrutinize the impact of the IMO 2020 on the liner shipping services and develop a goal programming model to optimize the investment decision of fleet upgrading and ship fleet deployment strategies under fuel cost ambiguity and demand uncertainty.
- We build a tangible mathematical model, called the ambiguous robustness optimization (ARnO) model, to account for the misspecification errors of the ambiguity set of the fuel cost. We define the financial riskiness index (FRI) from the ARnO and show a nice interpretation of ARnO from risk perspectives.
- We develop a solution scheme for ARnO model by showing that those constraints involving uncertainty in the ARnO model can be reformulated using exponential cones and solved efficiently through the mixed-integer second-order cone programming (MISOCP) approximation.

- We compare the payback time model, the goal programming model and ARnO model on a real problem of the Asia-North America West Coast (Pacific Northwest) shipping network operated by a world's leading ocean container carrier, evaluate our models' performance and draw rich managerial insights.

Overview of the paper

The paper unfolds as follows. We first give a concise literature review in §2. In §3, we introduce the liner ship fleet upgrading problem towards the IMO 2020. In §4, we build the goal programming model to maximize the expected NPV subject to the ship fleet capacity utilization constraints. In §5, we introduce the ARnO model to define the FRI and show that the developed model can be efficiently solved via an MISOCP approximation. Numerical examples are presented in §6 and §7 concludes the paper. Any proof not in the main text can be found in the Appendix.

Notations To facilitate our presentation, we adopt the following notations throughout the paper. We use the calligraphic font to denote a set such as $\mathcal{T}$ with its cardinality $|\mathcal{T}|$. We denote by $\mathbb{R}$, $\mathbb{R}_+$, and $\mathbb{B}$ the sets of real numbers, nonnegative real numbers, and binary integers, respectively. We use boldface lowercase letters to represent vectors. We use $\mathcal{P}_0(\mathcal{Z})$ to represent the set of all probability distributions on support set $\mathcal{Z}$. We use tilde $(\tilde{\cdot})$ to denote uncertain parameters and use $\mathbb{P}$ to denote probability measure on sample space Ω. We denote by $\mathbb{E}_{\mathbb{P}}[\tilde{\boldsymbol{z}}]$ the expectation of $\tilde{\boldsymbol{z}}$ under probability distribution $\tilde{\boldsymbol{z}} \sim \mathbb{P}$. The inequality between two uncertain parameters $\tilde{t} \geq \tilde{v}$ describes state-wise dominance, i.e., $\tilde{t}(\omega) \geq \tilde{v}(\omega)$ for all $\omega \in \Omega$. We denote by $\mathbb{Q} \ll \mathbb{P}$ that the probability measure $\mathbb{Q}$ is absolutely continuous with respect to measure $\mathbb{P}$.

2 Related Studies

In this section, we review the related studies in terms of methodology and model: methodology-wise, our study is concerned with decision-making under ambiguity; model-wise, we highlight the difference between our model and the traditional liner ship fleet renewal models.

2.1 Decision-making Under Ambiguity

Optimization under uncertainty relies on the model of uncertainty. If the uncertain parameters are assumed to be governed by a known probability distribution, stochastic programming can be applied to minimize the total average cost under the probability distribution (see, e.g., Birge and Louveaux 2011, Shapiro et al. 2014). When an uncertainty set is used to describe all the possible realizations of uncertain parameters, classic robust optimization (RO) aims to find a decision that hedges against the worst-case outcome within the uncertainty set and hence is pessimistically biased, i.e., the realized objective value by robust solution would often be better than the optimal

value of robust optimization problem (see, e.g., Ben-Tal et al. 2009). When the model of uncertainty is characterized by an ambiguity set containing all possible probability distributions of uncertain parameters, distributionally robust optimization (DRO) seeks a decision that hedges against the worst-case distribution within the ambiguity set (see, e.g., Delage and Ye 2010, Wiesemann et al. 2014). Note that RO is a particular case of DRO because an uncertainty set is a special ambiguity set containing only support information of uncertain parameters. Hence, DRO can reduce the conservativeness of RO by incorporating more information that is compatible with the decision-maker's prior information into ambiguity sets.

The aforementioned modeling frameworks assume a particular model of uncertainty. However, the model of uncertainty might be far away from reality. In view of potential misspecification, Long et al. (2019) proposes a robustness optimization model where the decision-maker specifies a target, or an acceptable loss of optimality compared with a baseline model, as a trade-off for the model's ability to withstand greater uncertainty. Long et al. (2019) also shows that robustness optimization is associated with the axiomatized decision criterion termed *fragility measure* that has the properties of monotonicity, positive homogeneity, subadditivity, pro-robustness, and anti-fragility. The fragility measure is closely related to the riskiness index of Aumann and Serrano (2008), which has wide applications, including project portfolio selection (Hall et al. 2015) and vehicle routing (Jaillet et al. 2016), etc.

In terms of methodology, our studies propose a robustness optimization model that guides decision-making under an ambiguity set that describes the uncertain parameters. This is to be contrasted with Long et al. (2019), where a nominal distribution is assumed. Hence, we adapt their robustness optimization modeling framework to ambiguous distributions and define the FRI to evaluate the violation risk of a solution in meeting an NPV target under ambiguity. We optimize over FRI to mitigate financial risk.

2.2 The LSFU versus the Liner Ship Fleet Renewal (LSFR) Problem

The LSFR problem is the liner ship counterpart of the general fleet renewal problem (FRP), a family of problems in the Operations Research (OR) literature that studies the renewal and adaption of the existing fleet. The LSFR problem is also a subset of maritime fleet renewal problem (MFRP) that is concerned with the strategic level decision regarding: (i) the types of ships to add to the fleet or dispose of and (ii) when and how to do so to satisfy the transportation demand at the minimum cost (Pantuso et al. 2016). The LSFR can be seen as a dynamic variant of the fleet size and mix problem (FSMP), which is concerned with the strategic level decisions regarding the type and number of ships that an ocean carrier keeps in its ship fleet (Meng et al. 2014). Both FSMP and LSFR are usually investigated in combination with the tactical decision of fleet deployment. Readers

interested in fleet deployment problems and other OR applications in maritime transportation can refer to Ronen 1993, Christiansen et al. 2004, Christiansen et al. 2013 and Meng et al. 2014 for a comprehensive review representing the literature in the last three decades.

It is worth emphasizing that our model contrasts with the traditional LFRP models in the following three aspects. First, we study the LSFU in response to regulatory changes in the form of the IMO emission cap under uncertain fuel cost with a partially characterized probability distribution. Fleet upgrading decisions concerning when and how to invest in each vessel in an existing fleet are optimized to maximize the NPV of upgrading investment. In this regard, the traditional ship fleet renewal decisions, such as acquisition, selling, scrapping, chartering in and out, are usually pinned down before the LSFU decisions. Second, the fleet deployment problem in our model is meant for supporting the vessel retrofitting plan while ensuring limited interruption to the existing services through better fleet assignment, so we take the routing and sailing speed as given parameters according to the *status quo* other than decisions to make as in the tradition MFRP literature. Third, one key feature of our model is its ability to mitigate the financial risk from uncertain fuel cost, which, we believe, is an important strategic issue behind any upgrading investment but has been remained silent in the traditional MFRP. This is evidenced by Balland et al. (2015) who conducted a survey among internal stakeholders of a shipping company and found that risk is the most valued non-financial factor in air emission control investment.

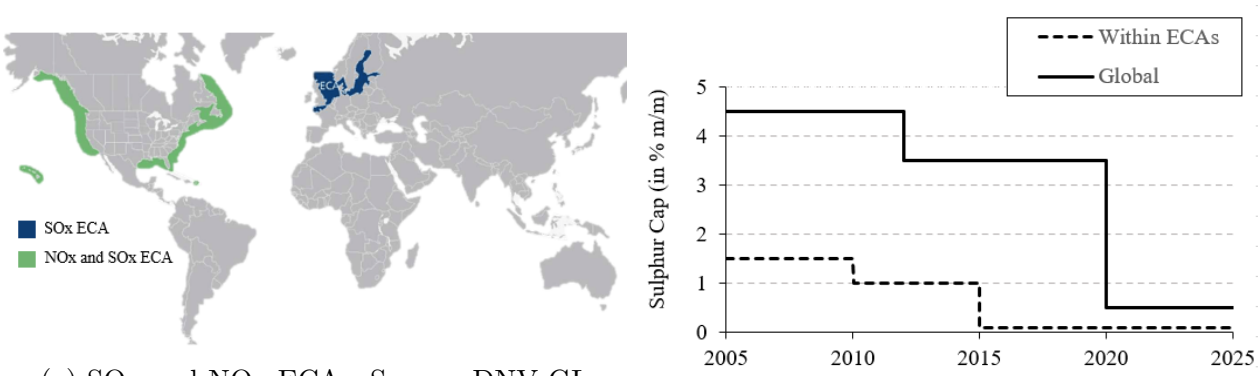
3 Problem Description

Recall that the main challenge behind the IMO 2020 compliance for an ocean carrier is to determine how to comply with the stringent emission cap and when to upgrade the existing ships, if necessary, to minimize cost while ensuring a minimum interruption to the existing services. In this section, we scrutinize and characterize the three widely recognized alternatives: (i) switching to VLSFO, (ii) installing EGCS, (iii) retrofitting vessels to use LNG, and describe the notations and assumptions in the LSFU problem. On account of the proceeding ship deployment issues raised from the fleet upgrading decisions, we elaborate the liner shipping service information. We also illustrate the calculation of payback time.

3.1 LSFU Towards IMO 2020

Increasingly stringent sulphur caps have been enforced under the auspices of IMO. Figure 2 illustrates the evolution of sulphur limits inside and outside the sulphur emission control areas (ECAs)[8]. The most recent regulation on the global sulphur cap, the IMO 2020, limits the sulphur content in ships' fuel oil by 0.5% m/m. There is an even stricter limit of 0.10% m/m in effect in ECAs.

[8] There are four established ECAs: the Baltic Sea area; the North Sea area; the North American area (covering designated coastal areas of the United States and Canada); and the United States Caribbean Sea area (around Puerto Rico and the United States Virgin Islands).



(a) SOx and NOx ECAs. Source: DNV GL

(b) The evolution of marine sulphur limit

Figure 2 Regulatory requirement according to International Convention for the Prevention of Pollution from Ships (MARPOL) Annex VI.

In this section, we characterize the three IMO 2020 solutions according to the dimensions parameterized in Table 1. The parameters in Table 1 will be explained in the subsequent discussions.

Table 1 Parameters for IMO Sulphur 2020, for any vessel k in period t

Compliance solutions	Fuel cost non-ECA	Fuel cost ECA	Lead time	Fixed cost	Loss of slots
Fuel switch	$\tilde{c}_{k,t}^{VLSFO}$	$\tilde{c}_{k,t}^{MGO}$	0	0	0
Install EGCS (also known as scrubbers)	$\tilde{c}_{k,t}^{HSFO}$	$\tilde{c}_{k,t}^{HSFO}$	τ_k^{EGCS}	FC_k^{EGCS}	η_k^{EGCS}
Retrofit vessels to use LNG	$\tilde{c}_{k,t}^{LNG}$	$\tilde{c}_{k,t}^{LNG}$	τ_k^{LNG}	FC_k^{LNG}	η_k^{LNG}

Planning horizon. Consider a long-term planning horizon $\mathcal{T} = \{1, ..., T\}$ for the LSFU where the length of a single period, denoted by Δt (in weeks), can be set according to the dynamics of demand forecasted within the planning horizon. For example, Δt can be a month, a year, etc.

Ship fleet composition. At the beginning of the planning horizon, a fleet of vessels operates in a particular liner shipping network, represented by set $\mathcal{K}$ and indexed by $k = 1, ..., K$. Associated with each vessel k, there is a vessel capacity V_k (in TEU) and an initial state indicator $(\alpha_{k,0}, \beta_{k,0})$ where $\alpha_{k,0} = 1$ if and only if the vessel has been converted to LNG, and $\beta_{k,0} = 1$ if and only if a scrubber has been installed to the vessel. Otherwise $(\alpha_{k,0}, \beta_{k,0}) = (0, 0)$.

Fuel cost. Different types of bunker fuels are required under the three choices. Without any investment, it is necessary to switch to the more expensive VLSFO and MGO outside and inside the ECA, respectively. Scrubbers allow operations using the regular HSFO. Let $\tilde{c}_{k,t}^{VLSFO}$, $\tilde{c}_{k,t}^{MGO}$, $\tilde{c}_{k,t}^{HSFO}$ and $\tilde{c}_{k,t}^{LNG}$ denote the bunker fuel cost (in USD per day) of vessel k under the four types of fuel in period t, respectively.

To determine the optimal LSFU strategy, it sounds tempting to predict the bunker fuel prices over the long run. In fact, our goal does not necessitate such price prediction since we are only

interested in the spreads (in USD per day) between any two alternative fuels. To simplify notation, we denote by $\tilde{\boldsymbol{\delta}} := (\tilde{\delta}^{VL}_{k,t}, \tilde{\delta}^{VH}_{k,t}, \tilde{\delta}^{MV}_{k,t})_{k\in\mathcal{K},t\in\mathcal{T}}$ the random vector of fuel cost spreads with

$$\tilde{\delta}^{VL}_{k,t} := \tilde{c}^{VLSFO}_{k,t} - \tilde{c}^{LNG}_{k,t}; \qquad \tilde{\delta}^{VH}_{k,t} := \tilde{c}^{VLSFO}_{k,t} - \tilde{c}^{HSFO}_{k,t}; \qquad \tilde{\delta}^{MV}_{k,t} := \tilde{c}^{MGO}_{k,t} - \tilde{c}^{VLSFO}_{k,t}. \tag{1}$$

To characterize the uncertainties in fuel cost spreads, we adopt a factor-based model (see, e.g., Ben-Tal et al. 2009, Hall et al. 2015).

Assumption 1 **(Factor-based ambiguity set of fuel cost spreads)**. *We assume a factor-based model of fuel cost spreads uncertainty as follows:*

$$\tilde{\boldsymbol{\delta}} = \left\{ (\tilde{\delta}^{VL}_{k,t}, \tilde{\delta}^{VH}_{k,t}, \tilde{\delta}^{MV}_{k,t})_{k\in\mathcal{K},t\in\mathcal{T}} \;\middle|\; \begin{array}{l} \tilde{\delta}^{VL}_{k,t} = \omega_k \Delta\tilde{c}^{VL}_t, \;\; \Delta\tilde{c}^{VL}_t = (\phi_{11}\tilde{\nu}^1_t + \phi_{12}\tilde{\nu}^2_t + \phi_{13}\tilde{\nu}^3_t) \;\; \forall t\in\mathcal{T}, k\in\mathcal{K} \\ \tilde{\delta}^{VH}_{k,t} = \omega_k \Delta\tilde{c}^{VH}_t, \;\; \Delta\tilde{c}^{VH}_t = (\phi_{21}\tilde{\nu}^1_t + \phi_{22}\tilde{\nu}^2_t + \phi_{23}\tilde{\nu}^3_t) \;\; \forall t\in\mathcal{T}, k\in\mathcal{K} \\ \tilde{\delta}^{MV}_{k,t} = \omega_k \Delta\tilde{c}^{VH}_t, \;\; \Delta\tilde{c}^{MV}_t = (\phi_{31}\tilde{\nu}^1_t + \phi_{32}\tilde{\nu}^2_t + \phi_{33}\tilde{\nu}^3_t) \;\; \forall t\in\mathcal{T}, k\in\mathcal{K} \end{array} \right\} \tag{2}$$

where ω_k is the fuel consumption scaling factor of vessel k, and ϕ_{ij}'s are the linear dependence coefficients of fuel price differences, i.e., $\Delta\tilde{c}^{VL}_t$, $\Delta\tilde{c}^{VH}_t$ and $\Delta\tilde{c}^{MV}_t$, on the independent uncertain factors $\tilde{\boldsymbol{\nu}}$ characterized by the ambiguity set $\mathcal{F} := \times_{t\in\mathcal{T},\ell\in[3]}\mathcal{F}^\ell_t$ with

$$\mathcal{F}^\ell_t := \left\{ \mathbb{P}\in\mathcal{P}_0(\mathbb{R}) \;\middle|\; \begin{array}{l} \tilde{\nu}^\ell_t \sim \mathbb{P} \\ \mathbb{E}_{\mathbb{P}}\left[\tilde{\nu}^\ell_t\right] = \hat{\nu}^\ell_t \\ \mathbb{E}_{\mathbb{P}}\left[|\tilde{\nu}^\ell_t - \hat{\nu}^\ell_t|\right] \le d^\ell_t \\ \mathbb{P}\left[\tilde{\nu}^\ell_t \in [\underline{\nu}^\ell_t, \overline{\nu}^\ell_t]\right] = 1 \end{array} \right\}. \tag{3}$$

Assumption 1 builds upon: (i) our empirical study on fuel prices (see §EC.3) and (ii) the perfect correlation between fuel cost spreads per vessel $\tilde{\boldsymbol{\delta}}_{k,t}$ and fuel price differences $\Delta\tilde{\boldsymbol{c}}_t$ via fuel consumption scaling factors ω_k that measures a vessel's bunker consumption rate in mmBtu[9] per day. Further, we assume correlation between fuel price differences and a set of mutually independent uncertain factors $\tilde{\boldsymbol{\nu}}$, all in unit of USD per mmBtu. For each $\tilde{\nu}^\ell_t$, we pin down its mean, support and mean absolute deviation (MAD) since they are intuitive to specify (see §EC.3.1). The MAD measures uncertainty's dispersion from its expected value. Compared to the standard deviation, it is less affected by outliers and considered more robust (see, e.g., Casella and Berger 2021). More distributional information might be incorporated while maintaining model tractability at the cost of more complicated ambiguity sets (see, e.g., Chen and Sim 2021). The uncertainty of $\tilde{\boldsymbol{\delta}}$ is fully characterizable by $\tilde{\boldsymbol{\nu}}$, so we also use $\tilde{\boldsymbol{\delta}} = \boldsymbol{\delta}(\tilde{\boldsymbol{\nu}})$ to show the dependence.

Assumption 2 **(Ambiguity set misspecification)**. *The ambiguity set $\mathcal{F}$ defined in Assumption 1 may be mis-specified.*

[9] The term "mmBtu" is short for Metric Million British Thermal Unit.

Assumption 2 means that it is possible that the distribution of $\tilde{\boldsymbol{\nu}}$, denoted by $\mathbb{Q}$, is not in $\mathcal{F}$. This is a natural assumption in the LSFU problem towards IMO 2020 since the planning horizon for upgrading investments span years or even decades, during which the energy markets are too volatile to be characterized accurately. As a counteraction to Assumption 2, we use robustness optimization to mitigate the potential misspecification.

Conversion period. Scrubber installation and vessel retrofitting have a long lead time. During this period, the vessel cannot operate and fulfill the targeted demand. Let τ_k^{EGCS} and τ_k^{LNG} denote the time needed for installation or retrofitting, respectively, for the vessel k.

Fixed cost. There are various types of scrubbers, with the "open-loop wet scrubbers" which requires the lowest cost to install and operate being the most used types (IHS Markit 2019). We assume that if EGCS is opted for vessel k, then an open-loop wet scrubber will be installed at a fixed cost denoted by FC_k^{EGCS}. Besides, denote by FC_k^{LNG} the fixed cost for LNG retrofitting. In practice, $FC_k^{EGCS} < FC_k^{LNG}$.

Loss of shipping slots. Scrubbers are usually several stories tall, which takes cargo space. LNG retrofitting takes even more space. For example, the pilot project by the carrier Hapag-Lloyd that converted a 15,000-TEU vessel, named Sajir, to be LNG-ready sees this big challenge where the fuel tank takes up 300 TEU slots [10]. Let η_k^{EGCS} and η_k^{LNG} denote the wastes of slots (in TEUs).

3.2 Liner Shipping Service Network

Next, we explain the notation and assumptions in the liner shipping services. We consider a fleet of heterogeneous vessels operated by an ocean carrier on a shipping network consisting of a set of predetermined routes where transshipment is permitted. The liner shipping company follows the existing established schedule, that is, the liner ship routes, sailing speed and service frequency are fixed according to the service schedule.

The fleet of vessels is operated according to a predetermined liner ship topology, coded as a 4-tuple $< \mathcal{R}, \mathcal{P}, \mathcal{W}, \mathcal{H} >$. The notation $\mathcal{R}$ represents the set of liner ship routes. Each route $r \in \mathcal{R}$ consists of a sequence of ports, denoted by set $\mathcal{P}_r$, that forms a directed loop represented by:

$$p_r^1 \to p_r^2 \to ... \to ... \to p_r^{m_r} \to p_r^1 \tag{4}$$

where $p_r^l \in \mathcal{P}_r$ represents the lth port of call in route r and m_r is the total number of ports of call on route r. Let $\mathcal{P}$ denote the set of all ports of call (among all routes), that is, $\mathcal{P} := \cup_{r \in \mathcal{R}} \mathcal{P}_r$. The voyage between any two consecutive ports along a particular liner ship route r forms a leg, and let $\mathcal{L}_r$ denote the set of all legs covered by the route r, indexed by $l = 1, ..., m_r$, such that: for

[10] `https://www.rivieramm.com/news-content-hub/news-content-hub/hapag-lloyd-behind-the-worldrsquos-largest-box-ship-lng-retrofit-55803`, accessed July 9, 2020

$l = 1, ..., m_r - 1$, leg l refers to the edge between p_r^l and p_r^{l+1}; and for $l = m_r$, leg l refers to the edge between $p_r^{m_r}$ and p_r^1. Let $\bar{t}_r$ denote the total sailing time of the route r, within which $\bar{t}_r^{ECA}$ is the sailing time that falls into the ECA region.

The notation $\mathcal{W}$ represents a set of origination-destination (O-D) pairs, and it is indexed by (o, d) where $o, d \in \mathcal{P}$. Associated with each $(o, d) \in \mathcal{W}$, we denote by $Q_t^{(o,d)}$ the transportation demand from the port of origin o to the port of destination d in period t. To predict $Q_t^{(o,d)}$ accurately over a long planning horizon is impracticable. Any future realization of demand is likely to deviate from the estimate, denoted by $\hat{Q}_t^{(o,d)}$. We adopt a robust optimization approach with a widely-used budget uncertainty set (Bertsimas and Sim 2004) to address demand uncertainty. We will discuss its benefits through our numerical experiments in §6.

Assumption 3. *We assume for each $t \in \mathcal{T}$, the vector of uncertain demand $\boldsymbol{Q}_t := \{Q_t^{(o,d)}\}_{(o,d)\in\mathcal{W}}$ lies in the uncertainty set*

$$\mathcal{Q}_t = \left\{ \boldsymbol{Q} \in \mathbb{R}_+^{|\mathcal{W}|} \;\middle|\; \begin{array}{l} Q_t^{(o,d)} = \hat{Q}_t^{(o,d)} + \Delta Q_t^{(o,d)}, \underline{\Delta Q}_t^{(o,d)} \le \Delta Q_t^{(o,d)} \le \overline{\Delta Q}_t^{(o,d)}, \forall (o,d) \in \mathcal{W}, \\ \sum_{(o,d)\in\mathcal{W}} |\Delta Q_t^{(o,d)}| \le \Gamma_t \end{array} \right\}$$

where $\underline{\Delta Q}_t^{(o,d)} \le 0$, $\overline{\Delta Q}_t^{(o,d)} \ge 0$, and $\Gamma_t \ge 0$ for each $(o, d) \in \mathcal{W}$, $t \in \mathcal{T}$.

Assumption 3 allows uncertain demand $Q_t^{(o,d)}$ to deviate from its estimate $\hat{Q}_t^{(o,d)}$ by $\Delta Q_t^{(o,d)}$ bounded in the interval $\left[\underline{\Delta Q}_t^{(o,d)}, \overline{\Delta Q}_t^{(o,d)}\right]$, with the sum of total demand absolute deviation bounded by Γ_t. When $\Gamma_t = 0$, $\mathcal{Q}_t$ degenerates to a singleton and there is no demand deviation. Larger Γ_t implies more protection against demand uncertainty. Thus, the choice of Γ_t is subjected to our belief about total deviation and degree of ambiguity aversion.

For each $(o, d) \in \mathcal{W}$, the ocean carrier can predetermine a set of candidate paths, denoted by $\mathcal{H}^{(o,d)}$ and indexed by $h^{(o,d)}$. Each path covers one or more than one leg that may come from the same route or several routes. Note that, if transshipment is allowed, each path $h^{(o,d)} \in \mathcal{H}^{(o,d)}$ may consist of either (i) legs from the same route or (ii) legs from different routes; in contrast, if transshipment is not allowed, (ii) is prohibited. Let $\rho_{r,l}^{h^{(o,d)}}$ be an indicator of whether a leg l of route r will be covered during the voyage of path $h^{(o,d)}$. We denote $\mathcal{H} = \cup_{(o,d)\in\mathcal{W}} \mathcal{H}^{(o,d)}$. An example of such a liner shipping network is illustrated in §6.1.

Assumption 4 **(Service status quo)**. *An ocean carrier preserves the services routes and sailing speed according to the current service schedule.*

This is a natural assumption due to the nature of liner shipping services where liner shipping companies publish their service routes, with fixed sequences of ports of call at a regular service frequency, to attract cargo (Meng et al. 2014).

3.3 Payback Time Model

One of the most straightforward investment appraisal approaches is the payback period—the length of time for an investment to break even. This approach often guides existing decision-making processes in practice.

We illustrate this approach of the payback period as a benchmark. Taking fuel switching as a default option, in period t, for any vessel k that is deployed on route r, under the expected fuel cost spreads explained in Assumption 1, the minimum payback times for scrubber installation and LNG retrofitting, respectively, are

[PP]

$$\begin{aligned} PP^{EGCS}_{r,k,t}(\tilde{\boldsymbol{\delta}}) &:= \min\left\{i : FC^{EGCS}_k < \textstyle\sum_{s=t+\tau^{EGCS}_k}^{t+i}(\tilde{\delta}^{VH}_{k,s}\bar{t}_r + \tilde{\delta}^{MV}_{k,s}\bar{t}^{ECA}_r)N_r\right\} \\ PP^{LNG}_{r,k,t}(\tilde{\boldsymbol{\delta}}) &:= \min\left\{i : FC^{LNG}_k < \textstyle\sum_{s=t+\tau^{LNG}_k}^{t+i}(\tilde{\delta}^{VL}_{k,s}\bar{t}_r + \tilde{\delta}^{MV}_{k,s}\bar{t}^{ECA}_r)N_r\right\}, \end{aligned} \tag{5}$$

where N_r is the number of loops covered per period per vessel under the existing weekly service of route r. The RHS of the inequality in Model (5) accounts for an investment's benefits in terms of fuel costs saved. We use notation $PP^{EGCS}_{r,k,t}(\tilde{\boldsymbol{\delta}})$, $PP^{LNG}_{r,k,t}(\tilde{\boldsymbol{\delta}})$ to represent the dependence between the minimum payback times and random fuel cost $\tilde{\boldsymbol{\delta}}$.

At any time t, given knowledge of $\tilde{\boldsymbol{\delta}}$, the retrofitting decision, either scrubber installation or LNG retrofitting of the vessel k on route r, is evaluated by comparing the minimum payback time, $PP^{EGCS}_{r,k,t}$ or $PP^{LNG}_{r,k,t}$, with the planning horizon T. An investment will not be considered unless $T-t$ exceeds or is equal to the minimum payback time, that is, $PP^{EGCS}_{r,k,t} \leq T-t$ or $PP^{LNG}_{r,k,t} \leq T-t$.

4 The Mathematical Model

This section is concerned with model formulation to mitigate financial risk for the LSFU towards the IMO 2020. We use two criteria for fleet upgrading decisions: (i) the NPV of any investment (in scrubber installation or LNG retrofitting) should be maximized (ii) the interruption to existing services in terms of demand fulfillment should be minimized. These two criteria and the corresponding mathematical models will be explained in detail in the subsequent two sections.

To bring forward the model, we first define the decision variables associated with ship fleet upgrading. Let $\alpha_{k,t}$ and $\beta_{k,t}$ be binary decision variables such that $\alpha_{k,t}$ equals 1 if and only if vessel k has been sent for LNG conversion in period t, and similarly, $\beta_{k,t}$ equals 1 if and only if vessel k has been sent for scrubber installation in period t. Note that if a vessel k has been modified in period $t-1$, it must have been modified too in period t, that is,

$$\alpha_{k,t} \geq \alpha_{k,t-1} \text{ and } \beta_{k,t} \geq \beta_{k,t-1} \qquad \forall k \in \mathcal{K}, t \in \mathcal{T} \tag{6}$$

where $\alpha_{k,0}$ and $\beta_{k,0}$ are the initial state of vessel k explained in §3.1. Assume that each vessel will be modified at most once within the planning horizon. For example, if any vessel has installed a scrubber, it will not be retrofitted to use LNG anymore, and vice versa. Then, we have

$$\alpha_{k,t} + \beta_{k,t} \leq 1 \qquad \forall k \in \mathcal{K}, t \in \mathcal{T}. \tag{7}$$

While the binary decision variables defined above characterize the choices among the three alternatives, we introduce another two sets of decision variables related to shipping deployment: $x_{r,k,t} \in \mathbb{B}$ to capture whether vessel k is assigned to route r in period t and $z_t^{h^{(o,d)}} \in \mathbb{R}_+$ to pin down the number of containers served through path $h^{(o,d)}$ in period t. We relax the integer requirement for $z_t^{h^{(o,d)}} \in \mathbb{R}_+$ as it is a big number, usually greater than 1000 (Meng and Wang 2011). Note that any vessel can not be assigned and operated in more than one route simultaneously, namely,

$$\textstyle\sum_{r \in \mathcal{R}} x_{r,k,t} \leq 1 \qquad \forall k \in \mathcal{K}, t \in \mathcal{T}. \tag{8}$$

Decision Criterion 1: Maximizing Net Present Value (NPV)

Whether any investment in the EGCS or LNG retrofitting provides a more profitable solution than the fuel switch depends on the trade-offs between the fixed cost incurred and the fuel cost saved. We model such trade-off through the NPV for investment. In particular, we consider the fixed scrubber installation cost and LNG retrofitting cost, if any, together with the benefits from such installation or retrofitting in the form of fuel costs saved. The fixed cost incurred in period t is

$$FC_t := \sum_{k \in \mathcal{K}} FC_k^{LNG}(\alpha_{k,t} - \alpha_{k,t-1}) + FC_k^{EGCS}(\beta_{k,t} - \beta_{k,t-1}). \tag{9}$$

To see Equation (9), $\alpha_{k,t} - \alpha_{k,t-1} = 1$ if and only if LNG retrofitting has been initiated in period t, and takes the value of 0 otherwise. Similarly, $\beta_{k,t} - \beta_{k,t-1} = 1$ if and only if the vessel is sent to install scrubber in period t.

Next, we consider voyage costs saved. Based on the fuel prices within and outside the ECA summarized in Table 1, the fleet's fuel cost saved in a period t due to any technology investment is a random variable depending on the realization of fuel cost spreads, as expressed below. For more details on the mathematical derivation of this expression, readers may refer to §EC.2.1.

$$\begin{aligned} CS_t(\tilde{\boldsymbol{\delta}}) := \sum_{k \in \mathcal{K}} \sum_{r \in \mathcal{R}} \{ & \tilde{\delta}_{k,t}^{VL} \bar{t}_r x_{r,k,t}(\alpha_{k,t} - \alpha_{k,0}) + \tilde{\delta}_{k,t}^{VH} \bar{t}_r x_{r,k,t}(\beta_{k,t} - \beta_{k,0}) \\ & + \tilde{\delta}_{k,t}^{MV} \bar{t}_r^{ECA} x_{r,k,t}(\alpha_{k,t} + \beta_{k,t} - \alpha_{k,0} - \beta_{k,0}) \} N_r. \end{aligned} \tag{10}$$

As such, the NPV of the upgrading investment, namely $\{(\alpha_{k,t}, \beta_{k,t})\}_{k \in \mathcal{K}, t \in \mathcal{T}}$, can be calculated as follows.

$$NPV(\tilde{\boldsymbol{\delta}}) := \sum_{t=1}^{T} \gamma^t [CS_t(\tilde{\boldsymbol{\delta}}) - FC_t], \tag{11}$$

where γ is a discount factor to reflect the time value of money. For example, weight average cost of capital (WACC) is often used as a discount factor.

The $NPV(\cdot)$ defined in Equation (11) models the trade-offs between the fixed cost for extra investment and the fuel costs saved under the corresponding investment as a function of uncertain fuel cost spreads. We further assume the decision criterion of rational decision-makers as follows.

Assumption 5. *A rational risk-neutral decision-maker will make investment decisions on fleet $\mathcal{K}$, corresponding to $\{(\alpha_{k,t}, \beta_{k,t})\}_{k\in\mathcal{K},t\in\mathcal{T}}$, with the objective of maximizing expected NPV under ambiguity.*

Decision Criterion 2: Maximizing Capacity Utilization Rate (CUR)

Capacity utilization measures the percentage of TEU slots utilized. It is an essential determinant of an ocean carrier's profitability (Agarwal and Ergun 2008). Compared with other possible measurements such as revenue and profit margin, CUR is easily obtainable and more robust to unpredictable factors, e.g., freight rate, port dues and foreign exchange rates (Wang et al. 2013). It is defined as a weighted average number of containers transported per route, or

$$U := \sum_{(o,d)\in\mathcal{W}} g^{(o,d)} r^{(o,d)}, \tag{12}$$

where $r^{(o,d)}$ is the number of weekly laden containers (in TEUs) transported from port o to port d, and $g^{(o,d)}$ is a pre-determined non-negative coefficient associated with each O-D pair. Different values assigned to $g^{(o,d)}$'s yield different definitions of liner ship route CUR. For more details on how to pin down the value of $g^{(o,d)}$ for $(o,d)\in\mathcal{W}$, refer to Wang et al. 2013. The number of laden containers transported from port o to port d weekly amounts to $\frac{\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} z_t^{h^{(o,d)}}}{\Delta t}$ on average.

Weekly CUR should be maximized to enlarge the profit margin, but the fleet deployment process is constrained by demand and vessel capacity. For any demand realization $\boldsymbol{Q}_t$, we define the maximal CUR in period t as

$$U_t(\boldsymbol{Q}_t) := \max_{\boldsymbol{z}_t} \quad \sum_{(o,d)\in\mathcal{W}} g^{(o,d)} \left\{ \frac{\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} z_t^{h^{(o,d)}}}{\Delta t} \right\} \tag{13a}$$

$$\text{s.t.} \quad \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} z_t^{h^{(o,d)}} \le Q_t^{(o,d)} \qquad \forall (o,d)\in\mathcal{W} \tag{13b}$$

$$\sum_{k\in\mathcal{K}} (V_k - \eta_k^{LNG}\alpha_{k,t} - \eta_k^{EGCS}\beta_{k,t}) x_{r,k,t} N_r \ge \sum_{(o,d)\in\mathcal{W}} \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \rho_{r,l}^{h^{(o,d)}} z_t^{h^{(o,d)}} \quad \forall r\in\mathcal{R}, l=1,...,m_r \tag{13c}$$

$$z_t^{h^{(o,d)}} \ge 0 \qquad \forall h^{(o,d)}\in\mathcal{H}^{(o,d)},\ (o,d)\in\mathcal{W}. \tag{13d}$$

On the one hand, the total number of containers that can be delivered from port o to port d through any possible container path in a particular period t, or $\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} z_t^{h^{(o,d)}}$, cannot exceed

the demand, see Equation (13b). On the other hand, the total number of containers carried by vessel k on any leg l will not exceed the capacity of the vessels, see Equation (13c), where the LHS models the capacity net off against any waste slots. We handle the bi-linear terms in Equations (10) and (13c) by standard linearization techniques (see §EC.2.2).

Meanwhile, as explained in §3.1, it takes τ_k^{LNG} periods to retrofit vessels to use LNG. During this period of time, the vessel cannot operate. Hence, for any $k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R}$,

$$x_{r,k,t} = 0 \text{ if } \sum_{s=1}^{t} \alpha_{k,s} \in [1, \tau_k^{LNG}], \tag{14}$$

where $\sum_{s=1}^{t} \alpha_{k,s}$ refers to the time that has been spent for retrofitting up to period t. Similarly,

$$x_{r,k,t} = 0 \text{ if } \sum_{s=1}^{t} \beta_{k,s} \in [1, \tau_k^{EGCS}]. \tag{15}$$

We reformulate constraints (14) and (15) into linear constraints by introducing auxiliary decision variables (see §EC.2.3).

Next, we derive the following proposition as a necessary condition for the two decision criteria. We will take advantage of this necessary condition to eliminate some dominated feasible solutions and simplify our computation.

Proposition 1. *Under the decision criteria of NPV maximization and CUR maximization, any investment, EGCS installation or LNG retrofitting, will not be initiated to vessel k after time t if the time left within the planning horizon $T-t$ is smaller than or equal to $\underline{PP}_{k,t}^{EGCS}$ or $\underline{PP}_{k,t}^{LNG}$, respectively, where*

$$\underline{PP}_{k,t}^{EGCS} := \max\left\{i : i \le PP_{r,k,t}^{EGCS}(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}})), \forall r \in \mathcal{R}, \tilde{\boldsymbol{\nu}} \sim \mathbb{P} \in \mathcal{F}\right\}$$
$$\underline{PP}_{k,t}^{LNG} := \max\left\{i : i \le PP_{r,k,t}^{LNG}(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}})), \forall r \in \mathcal{R}, \tilde{\boldsymbol{\nu}} \sim \mathbb{P} \in \mathcal{F}\right\}$$

are the greatest lower bound of payback time. Namely, for $t = 1, ..., T$,

$$t > T - \underline{PP}_{k,t}^{EGCS} \Longrightarrow \beta_{k,t} - \beta_{k,t-1} = 0 \tag{16}$$

$$t > T - \underline{PP}_{k,t}^{LNG} \Longrightarrow \alpha_{k,t} - \alpha_{k,t-1} = 0. \tag{17}$$

A Goal Programming Model

The bi-objective nature of the problem, explained in decision criterion 1 and 2, brings to light the following goal programming model, which maximizes Criterion 1 subject to a CUR target to be

achieved for Criterion 2:

[LSFU]

$$\begin{aligned} \max \quad & \inf_{\mathbb{P}\in\mathcal{F}} \mathbb{E}_{\mathbb{P}}\left[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] \\ \text{s.t.} \quad & U_t(\boldsymbol{Q}_t) \geq \underline{U}_t(\boldsymbol{Q}_t) && \forall t\in\mathcal{T}, \boldsymbol{Q}_t\in\mathcal{Q}_t \\ & \text{constraint sets: } (6)-(11), (13)-(15) \\ & \alpha_{k,t}, \beta_{k,t}, x_{r,k,t} \in \mathbb{B} \end{aligned} \tag{18}$$

We observe that the objective function is equal to the expected NPV under any $\mathbb{P}\in\mathcal{F}$, i.e., $\forall\mathbb{P}\in\mathcal{F}: \inf_{\mathbb{P}\in\mathcal{F}}\mathbb{E}_{\mathbb{P}}\left[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] = NPV(\boldsymbol{\delta}(\hat{\boldsymbol{\nu}}))$ because $NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))$ is linear in $\tilde{\boldsymbol{\nu}}$ and $\mathbb{E}_{\mathbb{P}}[\tilde{\boldsymbol{\nu}}] = \hat{\boldsymbol{\nu}}$ is given in $\mathcal{F}$. In each period $t\in\mathcal{T}$, we require the maximum utilization rate $U_t(\boldsymbol{Q}_t)$ to achieve at least a CUR target adaptive to demand realization,

$$\underline{U}_t(\boldsymbol{Q}_t) := \lambda\cdot\sum_{(o,d)\in\mathcal{W}} g^{(o,d)}\frac{Q_t^{(o,d)}}{\Delta t},$$

for any $\boldsymbol{Q}_t\in\mathcal{Q}_t$ where $\lambda\in(0,1)$ is a satisfactory level that can be pre-specified by the decision-maker. Note that the set of constraints (18) are adaptive robust optimization constraints, since the decision variables $z_t^{h^{(o,d)}}$'s in Equation (13) are adaptive to demand realization. To circumvent the intractability of adaptive robust optimization, we adopt affine recourse adaptation technique (see, e.g., Ben-Tal et al. 2004) by restricting

$$z_t^{h^{(o,d)}}(Q_t^{(o,d)}) = z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \tag{19}$$

to obtain tractable linear optimization approximations, see EC.2.5. Consequently, the LSFU model can be solved by a mixed integer linear programming approximation. Further, as discussed after Assumption 3, when the demand $\boldsymbol{Q}_t$ coincides with the estimated value $\hat{\boldsymbol{Q}}_t$, i.e., $\Gamma_t = 0$, the LSFU model degenerates to a deterministic model, which we refer to as the *deterministic LSFU model*.

Following the discussion of Assumption 2, we next consider the case when the specification of the ambiguity set $\mathcal{F}$ may not be accurate in the first place. To address this, we introduce a robustness optimization model in the following section.

5 A Robustness Optimization Model

In this section, we formulate a robustness optimization model under distribution ambiguity and design a solution scheme through conic programming reformulation. We first formulate the robustness optimization model under distribution ambiguity, the so-called ARnO model, where "ARnO" is an abbreviation for Ambiguous Robustness Optimization. Following the analysis of ARnO model, we define the FRI to characterize the risk of expected NPV not meeting a specific target under the distribution ambiguity. The ARnO can be efficiently solved by the off-the-shelf solvers through an MISOCP approximation, as we show in later sections.

5.1 The ARnO Model and the FRI

In this subsection, we formulate the ARnO model and formally define our FRI. Suppose $\tilde{\boldsymbol{\delta}} \sim \mathbb{Q}$. Following the Assumption 5, a rational risk-neutral decision-maker is supposed to make investment decisions to optimize $\mathbb{E}_{\mathbb{Q}}\left[NPV(\tilde{\boldsymbol{\delta}})\right]$. Recall that the objective value obtained by the LSFU model refers to the best expected NPV under any reference distribution $\mathbb{P}$ described in Assumption 1. If a reference distribution $\mathbb{P}$ is correctly specified, then $\mathbb{E}_{\mathbb{Q}}\left[NPV(\tilde{\boldsymbol{\delta}})\right] = \mathbb{E}_{\mathbb{P}}\left[NPV(\tilde{\boldsymbol{\delta}})\right]$. Nevertheless, as discussed after Assumption 2, this is not the case for the LSFU problem explored in the current study: the planning horizon lasts for years during which the energy markets are too volatile to be predicted accurately. Consequently, it is inevitably possible that $\mathbb{E}_{\mathbb{Q}}\left[NPV(\tilde{\boldsymbol{\delta}})\right] \neq \mathbb{E}_{\mathbb{P}}\left[NPV(\tilde{\boldsymbol{\delta}})\right]$ when $\mathbb{Q} \neq \mathbb{P}$. We measure the discrepancy between the distribution $\mathbb{Q}$ and the reference distribution $\mathbb{P}$ with Kullback-Leibler divergence (Kullback and Leibler 1951) and denote it as $D_{KL}(\mathbb{Q}||\mathbb{P})$. Note $D_{KL}(\mathbb{Q}||\mathbb{P}) \geq 0$ with equality holds only if $\mathbb{Q} = \mathbb{P}$. Kullback-Leibler divergence is widely used in OR literature as a measurement of the distance between two distributions (see, e.g., Lim and Shanthikumar 2007).

A tangible counteract to mitigate this bias is to set a target for $\mathbb{E}_{\mathbb{Q}}\left[NPV(\tilde{\boldsymbol{\delta}})\right]$, denoted as B, and bound the expected violation of target achievement, denoted as $B - \mathbb{E}_{\mathbb{Q}}\left[NPV(\tilde{\boldsymbol{\delta}})\right]$. In particular, we bound such expected violation of target achievement to be within the discrepancy $D_{KL}(\mathbb{Q}||\mathbb{P})$ multiplied by μ. Similar approaches appear in a large body of OR literature, e.g., *globalized robust optimization* (Ben-Tal et al. 2009, 2017). We define the resulting ARnO model as follows.

[ARnO]

$$
\begin{aligned}
\min\ & \mu \\
\text{s.t. } & \mathbb{E}_{\mathbb{Q}}\left[B - NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] \leq \mu D_{KL}(\mathbb{Q}||\mathbb{P}) && \forall \mathbb{Q} \ll \mathbb{P}, \forall \mathbb{P} \in \mathcal{F} \\
& U_t(\boldsymbol{Q}_t) \geq \underline{U}_t(\boldsymbol{Q}_t) && \forall t \in \mathcal{T}, \boldsymbol{Q}_t \in \mathcal{Q}_t \\
& \text{constraint sets: } (6)-(11), (13)-(15) \\
& \alpha_{k,t}, \beta_{k,t}, x_{r,k,t} \in \mathbb{B}, \mu > 0.
\end{aligned}
$$

To explain the ARnO model, we compare its difference with LSFU model, that is, the objective function and the first constraint. Instead of maximizing expected NPV, we set a target B for expected NPV and try to protect target achievement against uncertainty. Suppose we are given a reference distribution $\mathbb{P}$ while the underlying true distribution of $\tilde{\boldsymbol{\nu}}$ is $\mathbb{Q}$, if $\mathbb{P} = \mathbb{Q}$, i.e., there is no misspecification, we should obtain $\mathbb{E}_{\mathbb{Q}}\left[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] \geq B$ since $D_{KL}(\mathbb{Q}||\mathbb{P}) = 0$ and the target is achieved. Otherwise, we obtain $\mathbb{E}_{\mathbb{Q}}\left[B - NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] \leq \mu D_{KL}(\mathbb{Q}||\mathbb{P})$, that is, we can guarantee that the expected violation of target achievement $B - \mathbb{E}_{\mathbb{Q}}\left[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right]$ is bounded by the discrepancy $D_{KL}(\mathbb{Q}||\mathbb{P})$ multiplied by μ for any arbitrary true distribution $\mathbb{Q}$. Recall that the uncertainty $\tilde{\boldsymbol{\nu}}$ in ARnO is characterized through the ambiguity set $\mathcal{F}$ and we do not have the exact distribution of $\mathbb{P}$,

so we allow it to be any distribution in the ambiguity set $\mathcal{F}$. Clearly, a smaller μ is preferred since it enforces smaller violations towards target achievement and the goal of robustness optimization is to minimize the multiple μ. For each B, we determine the smallest μ for which the problem ARnO remains feasible. This measures how "likely" we can attain the performance target B.

We define the objective value of ARnO as the FRI as it is closely related to the riskiness index of Aumann and Serrano (2008), which is defined for a random variable $\tilde{z} \sim \mathbb{P}$ as $\psi[\tilde{z}] := \inf\{\mu > 0 : C_\mu[\tilde{z}] \geq 0\}$ in terms of the certainty equivalent under Constant Absolute Risk Aversion (CARA)

$$C_\mu[\tilde{z}] := \begin{cases} -\mu \ln \mathbb{E}_{\mathbb{P}}\left[\exp\left(\frac{-\tilde{z}}{\mu}\right)\right] & \text{if } \mu > 0 \\ \mathbb{E}_{\mathbb{P}}[\tilde{z}] & \text{if } \mu = +\infty \\ \operatorname{ess\,inf} \tilde{z} & \text{if } \mu = 0 \end{cases}.$$

The relation is clear from the constraint reformulation in Proposition 2 as follows.

Proposition 2. *The first constraint in ARnO model is equivalent to*

$$\inf_{\mathbb{P} \in \mathcal{F}} C_\mu\left[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))\right] \geq B. \tag{20}$$

This equivalence brings to light another interpretation of the FRI, that is, the ARnO model finds a solution that satisfies the following two criteria:

- Criterion 1: the worst-case certainty equivalent of NPV of investment should achieve a target of B, i.e., Equation (20). The concept of worst-case certainty equivalent was proposed by Gilboa and Schmeidler (1989) based on several axioms of preference (*monotonicity, transitivity, completeness, certainty-independence, continuity, uncertainty aversion*) under distribution ambiguity.
- Criterion 2: in any period t, the maximal CUR should achieve the target $\underline{U}_t(\boldsymbol{Q}_t)$ for any $\boldsymbol{Q}_t \in \mathcal{Q}_t$, i.e., Equation (18).

5.2 Tractability of ARnO

We next show that the ARnO model can be reformulated as a mixed-integer exponential cone program (MIECP) under reasonable assumptions, and then develop an MISOCP approximation to solve the problem. Note that approximating an exponential cone with a few second-order cones is a common practice in the literature (see, e.g., Zhu et al. 2020) owing to the computational power of mixed-integer programming solvers, such as Gurobi and CPLEX, in addressing MISOCP.

Since Equation (20) involves the worst-case certainty equivalent evaluation, we next derive its tractable reformulation as follows.

PROPOSITION 3. *The first constraint in ARnO has the following equivalent exponential cone reformulation:*

$$B+\sum_{t\in\mathcal{T}}\gamma^t\left(FC_t+\sum_{\ell\in[3]}\xi_t^\ell\right)\le 0 \tag{21a}$$

$$\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}\theta_{t,1}^\ell+\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\theta_{t,2}^\ell+\left(1-\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}-\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\right)\theta_{t,3}^\ell\le\mu \qquad \forall t\in\mathcal{T},\ell\in[3] \tag{21b}$$

$$\mu\exp\left(\frac{\underline{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right)\le\theta_{t,1}^\ell \qquad \forall t\in\mathcal{T},\ell\in[3] \tag{21c}$$

$$\mu\exp\left(\frac{\overline{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right)\le\theta_{t,2}^\ell \qquad \forall t\in\mathcal{T},\ell\in[3] \tag{21d}$$

$$\mu\exp\left(\frac{\hat{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right)\le\theta_{t,3}^\ell \qquad \forall t\in\mathcal{T},\ell\in[3] \tag{21e}$$

$$f_t^\ell=-\sum_{k\in\mathcal{K}}\sum_{r\in\mathcal{R}}\omega_k\left(\phi_{2\ell}\bar{t}_r u_{r,k,t}+\phi_{1\ell}\bar{t}_r w_{r,k,t}+\phi_{3\ell}\bar{t}_r^{ECA}(w_{r,k,t}+u_{r,k,t})\right) \qquad \forall t\in\mathcal{T},\ell\in[3] \tag{21f}$$

Note that the above constraints involving exponential functions can be reformulated using the exponential cone defined as,

$$\mathcal{C}_{\exp}:=\{(x_1,x_2,x_3)\mid x_1\ge x_2\exp(x_3/x_2),x_2>0\}\cup\{(x_1,0,x_3)\mid x_1\ge 0,x_3\le 0\}.$$

For example, constraint (21c) can be written as $(\theta_{t,1}^\ell,\mu,\underline{\nu}_t^\ell f_t^\ell-\xi_t^\ell)\in\mathcal{C}_{\exp}$.

We next introduce a second-order cone (SOC) approximation of exponential cone constraints so that the ARnO can be efficiently solved by the off-the-shelf MISOCP solver such as Gurobi (Gurobi Optimization, LLC 2020).

PROPOSITION 4. *The exponential cone constraint $(x_1,x_2,x_3)\in\mathcal{C}_{\exp}$ can be approximated by the following SOC constraints:*

$$\begin{array}{l}\exists\kappa_1,...,\kappa_m\\ \chi_1,...,\chi_n\in\mathbb{R}\end{array}\quad s.t.\quad\begin{cases}\chi_0=x_1\\ (\chi_i+x_2,2\chi_{i+1},\chi_i-x_2)\in\mathcal{Q}^3,i=0,...,n-1\\ \sum_{j=1}^m W_j\kappa_j=2^{-n}x_3\\ \left(\chi_n-(1+G_j)\kappa_j,\chi_n-2x_2-(1-G_j)\kappa_j,2\sqrt{G_j}\kappa_j\right)\in\mathcal{Q}^3,j=1,...,m\end{cases} \tag{22}$$

where W_j and G_j, $j=1,...,m$ are Gaussian quadrature weights and nodes respectively and $\mathcal{Q}^3$ is the three-dimensional second-order cone $\mathcal{Q}^3=\left\{(x_1,x_2,x_3):x_1\ge\sqrt{x_2^2+x_3^2}\right\}$. Moreover, for any $a>1$ and $\epsilon>0$, there exists (m,n) with $m+n=O(\sqrt{\ln(1/\epsilon)})$ such that:

1. *if $0<x_1/a<x_2<ax_1$ and $(x_1,x_2,x_3)\in\mathcal{C}_{\exp}$, then $(x_1,x_2,x_3+x_2\epsilon)$ is feasible to constraints (22).*
2. *if $0<x_1/a<x_2<ax_1$ and (x_1,x_2,x_3) is feasible to constraints (22), then $(x_1,x_2,x_3+x_2\epsilon)\in\mathcal{C}_{\exp}$.*

Based on Proposition 4, we can obtain an MISOCP approximation of ARnO since each exponential cone constraint has an SOC approximation. For example, the constraint (21c) can be approximated by

$$\begin{cases} \chi_{t,1}^{\ell,0} = \theta_{t,1}^{\ell} \\ \left(\chi_{t,1}^{\ell,i} + \mu, 2\chi_{t,1}^{\ell,i}, \chi_{t,1}^{\ell,i} - \mu\right) \in \mathcal{Q}^3, i = 0, ..., n-1 \\ \sum_{j=1}^{m} W_j \kappa_{t,1}^{\ell,j} = 2^{-n} \left(\underline{\nu}_t^{\ell} f_t^{\ell} - \xi_t^{\ell}\right) \\ \left(\chi_{t,1}^{\ell,n} - (1+G_j)\kappa_{t,1}^{\ell,j}, \chi_{t,1}^{\ell,j} - 2\mu - (1-G_j)\kappa_{t,1}^{\ell,j}, 2\sqrt{G_j}\kappa_{t,1}^{\ell,j}\right) \in \mathcal{Q}^3, j = 1, ..., m \end{cases}$$

6 Numerical Experiments

In this section, we illustrate how the risk-based LSFU model via ARnO can be modeled and solved through a case study based on the Asia-North America West Coast (Pacific Northwest) service of APL, one of the world's leading ocean carriers for container shipping. Two questions are answered and verified through this computational experiment:

(Q1) Comparing with the existing decision-making process guided by the calculation of payback periods, what is the value of considering the operating decisions of fleet deployment in fleet upgrading decisions towards IMO 2020 Compliance?

(Q2) What are the cost and benefit of considering the robust notion of FRI using ARnO in decision-making?

The Pacific Northwest service consists of four shipping routes, NP1, NP2, NP3 and NP4, as illustrated in Figure EC.3. We consider the set of four routes as $\mathcal{R}$. The detailed service information about each route, including frequency, number of vessels employed on each route (or fleet size), ports of call, and the transit time between any two ports are retrieved from the APL official website (Source: `http://www.apl.com/our-offerings/services`, accessed July 2020), as shown in Table EC.2. Twenty-four heterogeneous vessels are employed on the four routes, six vessels per route, in order to maintain the current weekly service provided that the total duration to finish each route amounts to six weeks. We consider these 24 vessels as our fleet $\mathcal{K}$. The information on each vessel can be found from the same website, including vessel capacity in TEU and the current route a particular vessel is employed to, as shown in Table EC.3. We assume an average speed of 24 knots in this experiment. This speed is observed from the published data.

All the mixed-integer programming problems in this paper are solved by Gurobi 9.0 (Gurobi Optimization, LLC 2020). We use $m = n = 3$ in our SOC approximation of exponential cone.

6.1 Data Generation

We consider a planning horizon of 240 weeks (≈ 5 years), divided into 20 periods, with the length of a single period being a quarter of a year (i.e., $\Delta t = 12$ weeks). A set of 24 vessels $\mathcal{K}$ are currently

employed on the four routes. Associated with each vessel k, there is a vessel capacity V_k as shown in Table EC.3 and an initial state $(\alpha_{k,0}, \beta_{k,0})$ assumed to be $(0,0)$ for all vessels, that is, we assume no installation of scrubbers or LNG retrofitting to any vessel. The fleet of vessel $\mathcal{K}$ are operated according to the predetermined route $\mathcal{R} = \{NP1, NP2, NP3, NP4\}$, with the sequence of ports $\mathcal{P}_r$ for each route r indicated in Table EC.2, where $N_r = \frac{12\text{weeks}}{42\text{days}} = 2$. Note that $\mathcal{W}$ and $\mathcal{H}^{(o,d)}$ are pre-determined by the shipping line company. In this experiment, we set the parameter value of $\mathcal{W}$ and $\mathcal{H}^{(o,d)}$ according to the procedure explained in EC.3.4.

We simulate the value of demand on any (o,d) pair according to leg demand that is randomly generated from the interval 0.1 to 1.0 times the capacity of the largest ship employed on the routes involved. Similar demand generating procedures are adopted in Agarwal and Ergun (2008) and such proportions are also used in Fagerholt (1999). The Demand size estimate on each (o,d) pair, namely $\hat{Q}^{(o,d)}$, are set by solving a network flow problem that maximizes the total number of containers carried subject to the constraints of leg flows (approximated by leg demands). We consider this set of demand, denoted by $\{\bar{Q}^{(o,d)}\}_{(o,d)\in\mathcal{W}}$, as a base case and scale it by different factors to reflect the variations of demand over time. We present the parameter settings related to scrubber installation and LNG retrofitting in EC.3.5.

Before discussing the two questions raised, we consider the deterministic LSFU model and explore how the expected NPV varies under different $\underline{U}_t(\hat{\boldsymbol{Q}}_t)$. Specifically, we set $\underline{U}_t(\hat{\boldsymbol{Q}}_t)$ as $\lambda \cdot \sum_{(o,d)\in\mathcal{W}} \hat{Q}_t^{(o,d)}$, where $\lambda \in [0,1]$, corresponding to the λ fraction of the overall demand in period t and study how the optimal NPV respond to λ in the LSFU model. Figure 3 depicts the Pareto optimal NPV under different λ. Users of this model can specify the target CUR level accordingly.

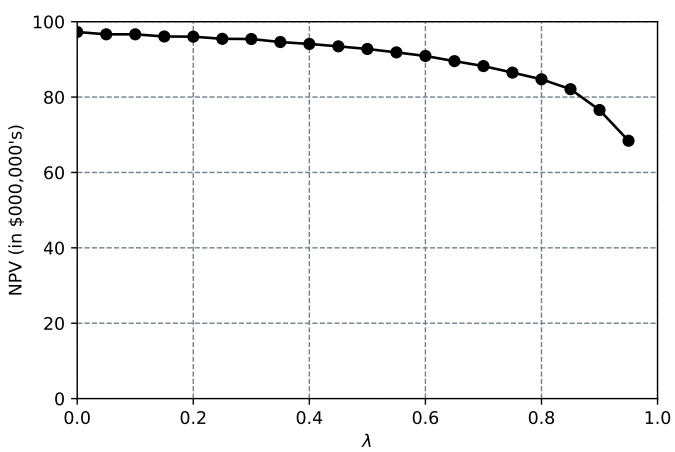


Figure 3 The Pareto optimal NPV under $\underline{U}_t(\hat{\boldsymbol{Q}}_t) := \lambda \cdot \sum_{(o,d)\in\mathcal{W}} \hat{Q}_t^{(o,d)}$ $(\Gamma_t = 0)$

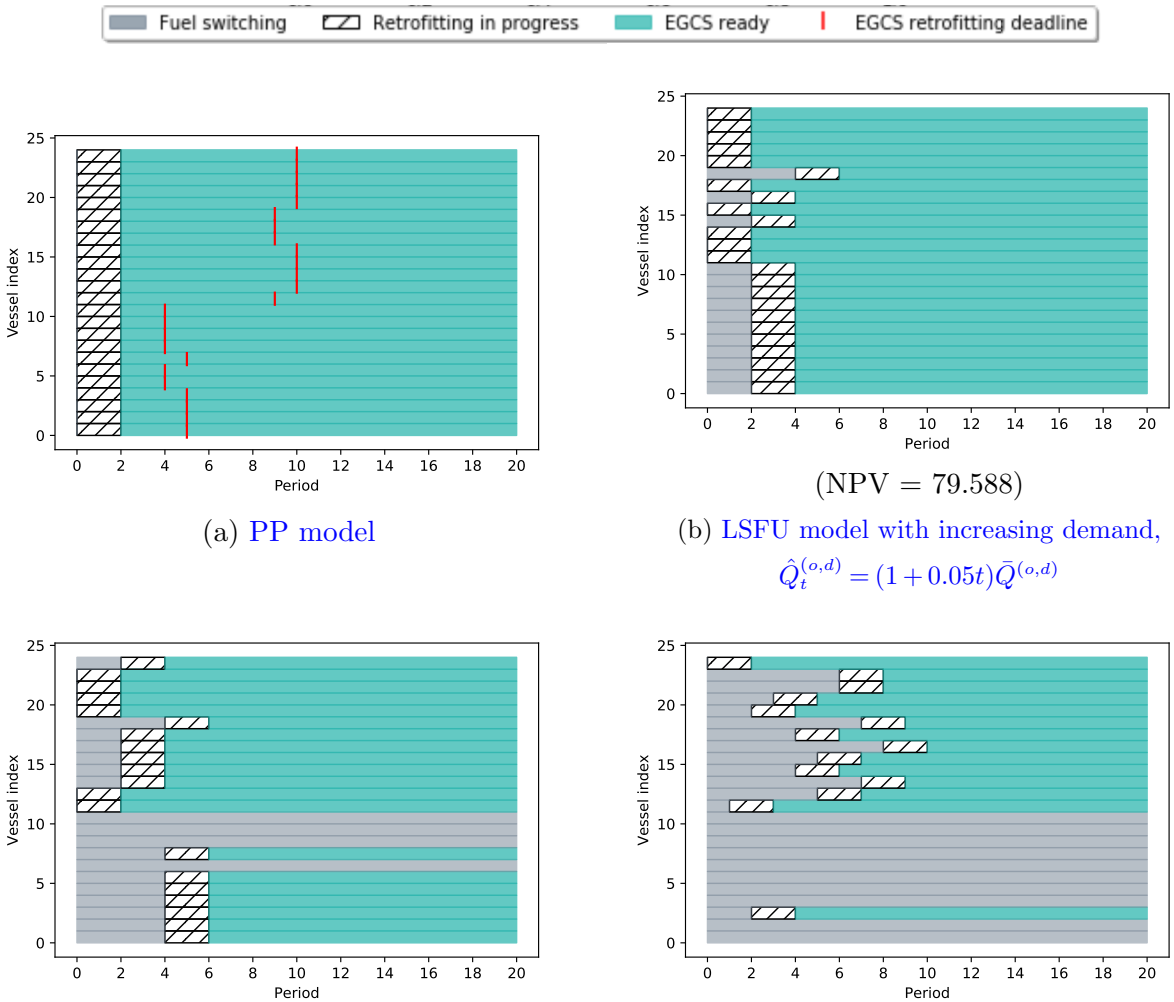
6.2 Q1: PP versus Deterministic LSFU Model

We first study the value of considering the operating decisions of fleet deployment in the fleet upgrading decisions towards IMO 2020 Compliance. In particular, we apply the mean value of uncertain factors $\tilde{\boldsymbol{\nu}}$ within the ambiguity set explained in Assumption 1, i.e., $\hat{\boldsymbol{\nu}}$, into payback time calculation (see model (5)). We compare the payback time with the result of a deterministic LSFU model.

The payback time for EGCS installation and LNG retrofitting are presented in the fourth and fifth column of Table EC.3, respectively. As observed from the result, among all the vessels that are assigned to any same route, the larger the vessel, the shorter the payback time, benefiting from the economies of scale. Under the given fuel price spread ambiguity set, it takes LNG retrofitting more than ten years to break even, which is beyond the planning horizon. In contrast, all vessels ought to be sent for EGCS installation because the payback time is within the length of the planning horizon. We visualize payback time for EGCS in Figure 4a. The red vertical segments show the deadlines to start EGCS retrofitting, after which an investment becomes unprofitable. If decision-makers ignore the operating level decisions of fleet deployment, investment should be made at the beginning as shown in the Gantt chart of Figure 4a in order to maximize the benefits from the retrofitting. Figure 4 presents the Gantt chart for fleet upgrading plan under PP model and LSFU model.

Next, we study the investment decisions guided by the deterministic LSFU models under a demand fulfillment target equivalent to 95% of the total demand over all (o,d) pairs in that period in terms of TEU, namely $\underline{U}_t(\hat{\boldsymbol{Q}}_t) = 0.95 \times \sum_{(o,d)\in\mathcal{W}} \hat{Q}_t^{(o,d)}$. To understand how the variations in trading capacity (or demand) over time affect investment decisions, we consider three cases: increasing, constant and decreasing demand. The aggregated demand over the planning horizon is controlled to be the same. The results are visualized in Gantt chart of Figures 4b, 4c and 4d. The horizontal axis indicates the number of periods, and the vertical axis refers to the vessel index. These three figures demonstrate the trade-offs between the profit maximization and capacity utilization. We summarize the major insights from the numerical results as follows:

- To maximize the NPV, EGCS installation should be made as early as possible, which is constrained by the capacity utilization requirement $U_t(\hat{\boldsymbol{Q}}_t) \geq \underline{U}_t(\hat{\boldsymbol{Q}}_t)$.
- The higher the initial demand in the first couple of periods, the greater the impact of such constraint, because EGCS installations limit the fleet deployment during the retrofitting lead time. Consequently, as we can tell by comparing the three figures, where Figure 4d visualizes the case with the highest level of demand initially and Figure 4b shows that of the lowest level of demand initially, fewer vessels are retrofitted in the first two periods in Figure 4d.



(a) PP model

(NPV = 79.588)

(b) LSFU model with increasing demand, $\hat{Q}_t^{(o,d)} = (1+0.05t)\bar{Q}^{(o,d)}$

(NPV = 68.871)

(c) LSFU model with constant demand, $\hat{Q}_t^{(o,d)} = [1+\frac{0.05(T-1)}{2}]\bar{Q}^{(o,d)}$

(NPV = 45.665)

(d) LSFU model with decreasing demand, $\hat{Q}_t^{(o,d)} = [1-0.05(t-T+1)]\bar{Q}^{(o,d)}$

Figure 4 Gantt chart and NPV (in \$000,000's) for fleet upgrading plan under PP model and LSFU model, $\underline{U}_t(\hat{\boldsymbol{Q}}_t) = 0.95 \times \sum_{(o,d)\in\mathcal{W}} \hat{Q}_t^{(o,d)}$, one period corresponds to three months

- The value of technology investments is also offset by the fulfillment of CUR targets. As shown in this numerical study, the higher the demand in the first couple of periods, the lower the NPV in technology investments.
- In addition, the total number of vessels scheduled for retrofitting decreases from twenty-four in Figure 4b to twenty in Figure 4c, and further to fourteen in Figure 4d. This shows that the retrofitting decisions must be carefully timed depending on the demand scenarios and cannot be determined in isolation.

In summary, the decision to retrofit the vessels depends crucially on the demand outlook in the near future, and the results indicate that fewer vessels should be retrofitted when demand is expected to decline steadily in the near future. According to the data from 2019 to 2021 shown

in Figure EC.4 from the website BIMCO[11], the pace of scrubber fittings has indeed slowed down since its peak in January 2020, as the emergence of the COVID-19 pandemic has eroded demand.

Compared with the LSFU model, the shortcoming of PP methods can be identified as follows. First, each vessel in the fleet is considered independently and her payback time is calculated according to the current route she is currently employed on. Secondly, the lead time and the waste of TEU slots are not quantified, potentially affecting capacity utilization. Third, the PP model can only provide answers to "how", but not "when", that is, it gives no instruction on when to send the vessel for retrofitting. Such retrofitting plans must be scheduled manually.

6.3 Q2: LSFU versus ARnO Model

Next, we discuss the benefits of the ARnO model over the LSFU model. We consider different target levels of NPV and CUR (i.e., the B and λ) and solve (i) the deterministic LSFU model (i.e., $\Gamma_t = 0, \forall t \in \mathcal{T}$) (ii) the robust LSFU model under demand uncertainty and (iii) the ARnO model. For models of (ii) and (iii), we construct a demand uncertainty set with $\underline{\Delta Q}_t^{(o,d)} = 0$ and $\overline{\Delta Q}_t^{(o,d)} = 0.2\hat{Q}_t^{(o,d)}$ to account for the risk of demand under-fulfilment due to capacity limitation, and we set $\Gamma_t = \sum_{(o,d)\in\mathcal{W}} \overline{\Delta Q}_t^{(o,d)} / \sqrt{|\mathcal{W}|}$ (to avoid Γ_t being too large or small). We implement the optimal solution of each model over four test data sets of cost spreads and demand realizations, each consisting of 1000 sample paths, and evaluate the NPVs by solving the lower level fleet deployment problem subject to the CUR target. If the problem is infeasible (i.e., achieving the CUR target is impossible), we return the maximum CUR attainable.

We simulate the sample path as follows. We first generate a random walk process $\tilde{Z}_t$ with independent increment $\tilde{X}_t \sim \text{Unif}(-1/3, 1/3)$. Then we simulate $\tilde{\boldsymbol{\nu}}_t$ as $\hat{\boldsymbol{\nu}}_t + \boldsymbol{d}_t \tilde{Z}_t$ (so we allow $\tilde{\boldsymbol{\nu}}_t$ to violate the independence in Assumption 1). Finally, we simulate $\tilde{\boldsymbol{\delta}}_t$ based on Equation (2) in Assumption 1 given the simulation results of $\tilde{\boldsymbol{\nu}}_t$. For demand uncertainty, we simulate $\tilde{\boldsymbol{Q}}_t$ as multivariate normal random variables. Each $\tilde{Q}_t^{(o,d)}$ has mean $\hat{Q}_t^{(o,d)}$ and standard deviation $0.3\overline{\Delta Q}_t^{(o,d)}$. The correlation coefficient between any two different demand $\tilde{Q}_t^{(o,d)}$ and $\tilde{Q}_t^{(o',d')}$ is equal to 0.2. To facilitate robustness check, we also consider $\tilde{Z}_t \sim \text{Unif}(-1/2, 1/2)$ with larger variance and larger demand correlation coefficient of 0.4. The distribution parameter of $\tilde{Z}_t$ such as 1/2 or 1/3 is chosen so that $\tilde{\boldsymbol{\delta}}$ is nonnegative with high probability, but in few extreme cases, the non-negativity may fail to hold.

We evaluate the NPVs according to five dimensions: mean, standard deviation, target achieving probability (TAP), value at risk (VaR) and conditional value at risk (CVaR) and report the CUR according its mean and TAP. The TAP is defined as the percentage times among the 1,000 sample

[11] `https://www.bimco.org/news/market_analysis/2021/20210429-high-sulphur-fuel-oil-sales-rebound-after-pre-imo-2020-correction`

Table 2 Performance of ARnO with different targets

Test data set[a]	Approach[b]	NPV (in k$)					CUR	
		Mean	Std[c]	TAP[d]	VaR[e]@95%	CVaR[f]@95%	Mean	TAP
V/12-Corr20	D-U85B100	77131.03	46672.70	47.10%	4958.37	1328.02	0.8373	8.80%
	R-U85B100	79255.70	47465.41	49.80%	6021.53	2556.46	0.8482	77.00%
	R-U85B85	78707.80	42784.34	58.50%	12715.89	9227.87	0.8492	88.50%
	R-U85B70	72748.91	33693.54	63.70%	20853.76	17798.40	0.8491	88.00%
	D-U95B100	64731.73	39686.70	48.10%	3634.36	946.94	0.9304	2.60%
	R-U95B100	62021.32	34820.39	48.90%	9167.83	6092.03	0.9423	34.90%
	R-U95B85	61130.05	31293.72	57.40%	13459.12	10712.95	0.9426	37.70%
	R-U95B70	61198.70	31273.51	64.30%	13563.10	10784.20	0.9425	37.00%
V/27-Corr20	D-U85B100	77094.84	38155.71	45.50%	14748.88	7092.43	0.8373	8.80%
	R-U85B100	79222.29	38829.52	49.70%	16479.24	8233.23	0.8482	77.00%
	R-U85B85	78675.76	34936.53	61.10%	21817.00	14622.29	0.8492	88.50%
	R-U85B70	72722.64	27493.40	69.90%	27907.34	22412.29	0.8491	88.00%
	D-U95B100	64711.15	32537.02	46.00%	11689.01	5279.00	0.9304	2.60%
	R-U95B100	62003.17	28575.57	47.50%	15608.09	9988.22	0.9423	34.90%
	R-U95B85	61110.08	25666.39	59.50%	19600.84	14396.75	0.9426	37.70%
	R-U95B70	61179.46	25642.51	70.80%	19668.99	14474.25	0.9425	37.00%
V/12-Corr40	D-U85B100	77070.06	46634.65	47.00%	5019.27	1386.19	0.8317	7.90%
	R-U85B100	78100.87	47080.82	49.20%	5573.63	2156.28	0.8443	58.40%
	R-U85B85	77598.49	42563.05	57.80%	12047.31	8718.87	0.8468	72.40%
	R-U85B70	71794.52	33389.45	63.10%	20548.89	17281.78	0.8470	75.80%
	D-U95B100	64648.06	39680.92	48.20%	3577.20	884.79	0.9177	0.60%
	R-U95B100	60811.85	34425.47	47.90%	7687.30	5478.26	0.9263	8.60%
	R-U95B85	59746.83	30820.03	56.00%	12333.71	10032.19	0.9266	9.80%
	R-U95B70	59843.89	30821.96	63.60%	12407.64	10119.91	0.9265	9.50%
V/27-Corr40	D-U85B100	77033.98	38112.40	45.50%	14794.37	7218.10	0.8317	7.90%
	R-U85B100	78070.88	38500.80	48.40%	14969.25	7792.19	0.8443	58.40%
	R-U85B85	77569.29	34794.61	59.70%	21432.37	14134.12	0.8468	72.40%
	R-U85B70	71776.57	27254.36	69.00%	27806.40	21970.82	0.8470	75.80%
	D-U95B100	64625.21	32530.41	46.00%	11716.75	5185.16	0.9177	0.60%
	R-U95B100	60793.32	28225.42	46.10%	14984.77	9282.84	0.9263	8.60%
	R-U95B85	59728.49	25246.45	57.80%	18533.27	13635.23	0.9266	9.80%
	R-U95B70	59825.99	25246.67	69.80%	18703.75	13728.18	0.9265	9.50%

[a.] V/X-Corr$Y$: V/X means the variance of $\tilde{Z}_t$ is $1/X$; Corr$Y$ means the demand correlation coefficients are Y%.
[b.] UXBY: $\underline{U}_t := X\% \cdot \sum_{(o,d)\in\mathcal{W}} Q_t^{(o,d)}$; $B := Y\% \times$[the objective value of LSFU model under UX]; when $Y = 100$, an ARnO model reduces to an LSFU model. A prefix "D" is refers to deterministic LSFU and a prefix "R" refers to robust LSFU approach.
[c.] Std = Standard deviation
[d.] TAP = Target achieving probability
[e.] VaR@β: $\beta-$Value at Risk
[f.] CVaR@β: $\beta-$Conditional Value at Risk.

paths, in which the NPV or CUR target is achieved. The value-at-risk (VaR) and conditional value-at-risk (CVaR) are two commonly used risk measures in financial risk management. They are defined with respect to a specific probability level β: the $\beta-$VaR measures the lowest amount α such that, with probability level β, the loss will not exceed α; the $\beta-$CVaR is the conditional expectation of losses above that amount α (Rockafellar et al. 2000). The results are shown in Table 2 and Figure EC.5. Note that the approaches with label containing "B100" correspond to the LSFU approach; prefix name "D" refers to deterministic LSFU; and "R" refers to robust LSFU or ARnO.

Several interesting results can be observed from this computational study. We discuss the cost and benefit of adopting the LSFU and ARnO models in addressing demand uncertainty and fuel cost uncertainty.

6.3.1 The robustness to demand uncertainty

Compared with the deterministic LSFU (refer to the approaches with labels "D" and "B100"), the robust LSFU models (approaches with labels "R" and "B100") bring significant improvement to the capacity utilization TAP. For example, the CUR's TAP under the R-U85B100 approach is almost nine times that under the D-U85B100 approach per the V/12-Corr20 data set. For another example, despite a substantial drop of realized capacity utilization when the CUR target increases from 85% to 95% (because a CUR target amounting to 95% of the total demand is harder to achieve), robust LSFU models demonstrate a much higher capacity utilization TAP than their deterministic counterparts. Such TAP improvements are less pronounced when the demands over all (o, d) pairs are more correlated (refer to the result of the "Corr40" test data set as compared with "Corr20" in Table 2), as a stronger correlation implies less demand pooling. Yet, we can adjust Γ_t to be larger to protect CUR against more significant demand variations.

We also observe that robust LSFU outperforms deterministic LSFU in TAP, VaR@95% and CVaR@95% of NPV. Further, a robust approach leads to a higher NPV mean and std than the deterministic approach under an 85% CUR level (D-U85B100 versus R-U85B100) but shows an opposite result in the case of 95% CUR level (D-U95B100 versus R-U95B100). We examine the underlying retrofitting decisions that drive the NPV differences. We find that under both models, it is optimal to upgrade all vessels in the cases of 85% CUR level. Still, the retrofitting decisions guided by a robust model are more likely to make the fleet deployment problem feasible to maximize the NPV. A retrofitting decision guided by the deterministic model often fails to achieve the CUR target and the objective will then switch to maximize capacity utilization. Hence, the retrofitting decisions under robust approaches attain higher NPV. In contrast, in the cases with a 95% CUR level, we

find that the robust solution (R-U95B100) retrofits fewer vessels than the deterministic solution (D-U95B100) to protect against the CUR target achievement. In that case, the deterministic approach leads to a higher NPV at the price of an extremely low capacity utilization TAP.

6.3.2 The value of ARnO model

Observe that, among all the test data sets, the NPV under the ARnO model demonstrates a substantial improvement in terms of risk (std, TAP, VaR@95% and CVaR@95%) at the cost of a slightly lower mean. For example, the boxplot of NPV in the test data set "V/12-Corr20" as visualized in Figure EC.5. Under this test data set, R-U85B85 improves the std of R-U85B100 by 4681.07 at the cost of 547.9 in terms of mean reduction. Similarly, ARnO improves VaR@95% and CVaR@95% significantly as well. We also observe that a lower NPV target B leads to lower NPV mean and std, and higher VaR@95% and CVaR@95% within each group of test data set. Notwithstanding higher expected NPVs demonstrated in the LSFU approach compared with those in the ARnO approach, it is owing to the correct specification of the expected value of cost spreads, a simulation assumption used in our experiment. This is very challenging in practice for similar reasons we explained after Assumption 2. Intuitively, the decision-maker sacrifices some expected NPV in exchange for a less risky outcome as NPV becomes less sensitive to the realization of uncertainties. When the variance of $\tilde{Z}_t$ increases (see the test data set with "V/12" versus "V/27"), the risk becomes higher, as observed from the increased std, decreased VaR@95% and CVaR@95% regardless of approaches. This risk can be mitigated by setting a lower NPV target B.

Tightening the feet deployment requirements through a higher CUR target weakens the trade-off between mean and std of NPV. For example, among all the data sets, under a CUR level of 85% (U85), changing the NPV target from 70% (B70) to 85% (B85) demonstrates a pronounced trade-off between mean and std of NPV; in contrast, under a CUR level of 95% (U95), increasing NPV targets by the same magnitude brings very tiny changes to mean and std of NPV. We examine the resulting investment decisions and find that the former (i.e., the pronounced trade-off) is accomplished by retrofitting more vessels. Holding up more vessels for retrofitting affects the operations and makes it infeasible for the carrier to fulfill the transportation demand under a CUR target as high as 95%.

Additionally, regardless of the approaches used and test data sets, a lower CUR target U leads to a higher expected NPV. This can be explained by the underlying trade-offs between the two criteria, similar to the trade-off visualized in Figure 3. Regarding the TAP of LSFU models, by the symmetry of $\tilde{Z}_t$, the TAP of NPV is around 0.5 among all test data sets. The lower the NPV target, the higher the TAP. Thus, compared with LSFU, ARnO has a lower NPV target and a higher TAP. Finally, we note that the NPV mean decreases as the demand correlation becomes

stronger (comparing the test data set under "Corr40" with that under "Corr20") for a similar reason discussed in §6.3.1. Stronger demand correlation across the service network destroys the risk pooling and makes achieving the CUR target more difficult. This distorts the objective from maximizing NPV to maximizing CUR whenever achieving the CUR target becomes infeasible.

7 Concluding Remarks

In this paper, we have studied a liner ship fleet upgrading problem towards the recently enforced sulphur emission cap, known as IMO 2020, under uncertain fuel prices characterized by ambiguity sets. The impact of IMO 2020 on liner shipping services is carefully reviewed and modeled, under which the ship upgrading plan (i.e., when and how to comply with the emission cap) is optimized in order to maximize the NPV of investments on a fleet of heterogeneous vessels while ensuring a minimum or limited interruption to the existing liner service operations in terms of fleet deployment. The resulting fleet deployment problem is concerned with CUR maximization under uncertain O-D pair demand profile where transshipment is allowed. To account for the potential misspecification of ambiguity set due to the long planning horizon and energy market volatility, we introduce an ARnO model that minimizes the FRI. This ARnO model turns out to be a mixed-integer exponential cone program that can be solved efficiently through MISOCP approximation. Our computational experiments suggest that the existing decision-making procedures guided by payback periods have some significant flaws. The LSFU model can improve the existing procedures. Also, the ARnO model can mitigate the financial risk with a slight sacrifice of expected NPV. This result brings to light some managerial implications in real applications: if the decision-makers are confident with the ambiguity set of the uncertain fuel price spread, then the LSFU model is a good enough approach of choice; in contrast, in many cases of practical importance where a precise specification of an ambiguity set is impossible, ARnO provides a better solution for mitigating the financial risk associated with emission cap compliance.

Future studies can build parallel models to study other maritime operations problems involving financial or operational risks due to the uncertainties in cargo demand (see e.g., Lee et al. 2021), empty container supply and demand (see e.g., Shu and Song 2014), freight rate, etc. For example, Assumption 4 can be relaxed to incorporate the tactical level decisions such as speed optimization into the LSFU optimization model to account for the relationship between sailing speed and bunker consumption rate. Other decision parameters in liner ship routing and scheduling can be considered in the underlying model in the same vein. More importantly, our method can be extended to a more general context of infrastructure, superstructure, and vehicle upgrade investment appraisal under uncertainty. We hope our work could encourage future studies on maritime transportation to address practically relevant challenges involving uncertainties from a risk mitigation angle.

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E-Companion

EC.1 Figures

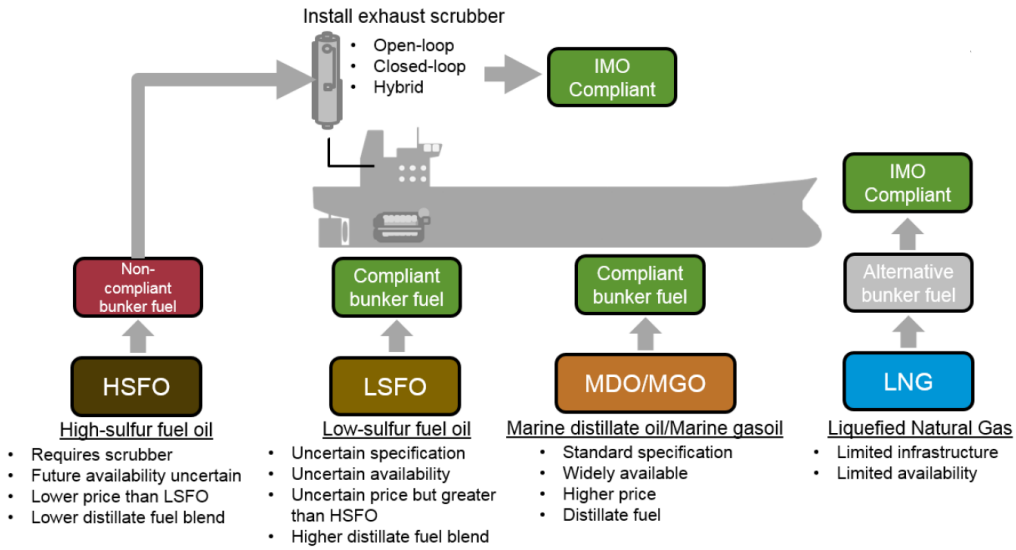


Figure EC.1 Shipping industry compliance pathways. Source: The U.S. Energy Information Administration (EIA): The Effects of Changes to Marine Fuel Sulfur Limits in 2020

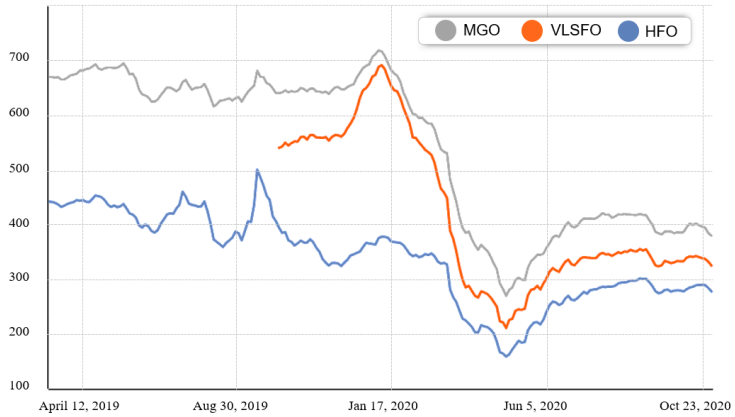


Figure EC.2 Bunker prices of the global 20 ports average from March 13, 2019 to October 13, 2020. Source: Ship & Bunker

Note that VLSFO is new fuel in response to the IMO 2020 obligation, so the price information before October 2019 is unavailable.

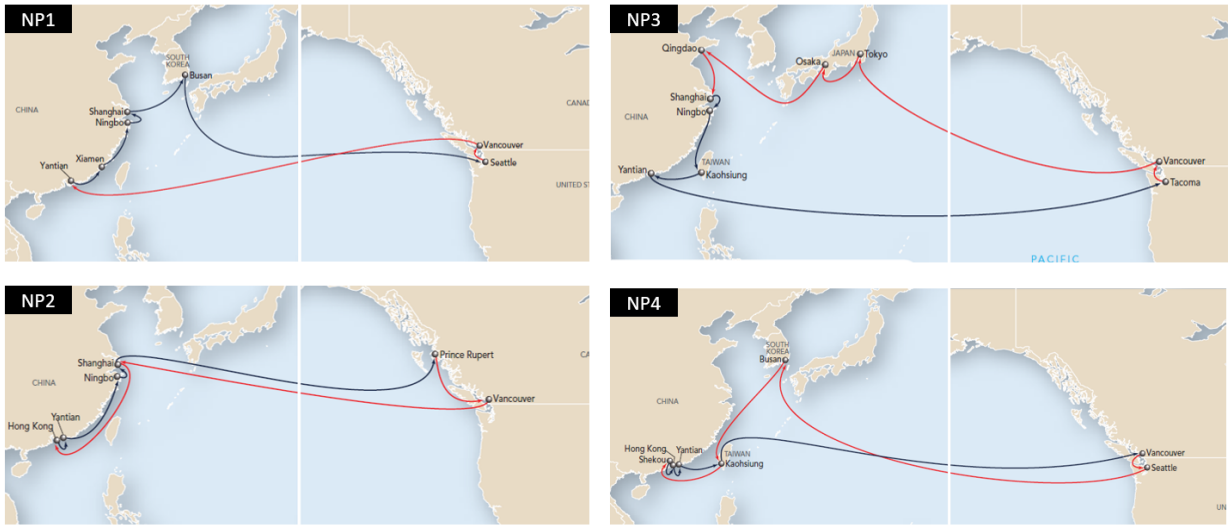


Figure EC.3 APL Asia-North America West Coast (Pacific Northwest) service

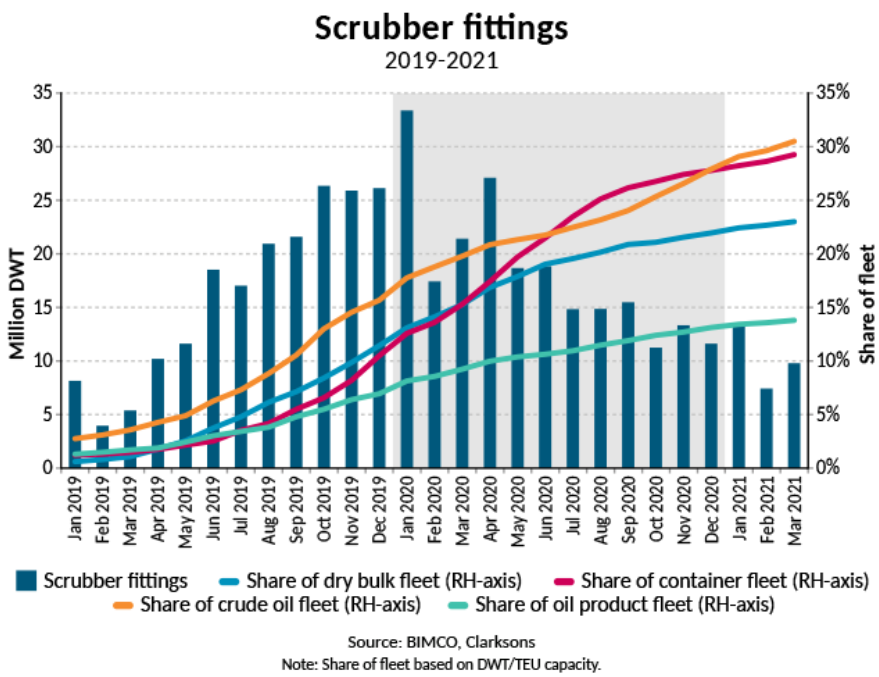


Figure EC.4 Share of bunker fuels in major bunkering hubs

EC.2 Proofs and reformulations

EC.2.1 Mathematical derivation of Equation (10)

Based on the fuel prices within and outside the ECA in Table 1, the fuel cost of vessel k in period t on route r can be estimated by

$$VC_{r,k,t}(\tilde{\boldsymbol{c}}) := \begin{cases} \tilde{c}_{k,t}^{VLSFO}(\bar{t}_r - \bar{t}_r^{ECA}) + \tilde{c}_{k,t}^{MGO}\bar{t}_r^{ECA} & \text{for } \beta_{k,t} = \alpha_{k,t} = 0 \\ \tilde{c}_{k,t}^{HSFO}\bar{t}_r & \text{for } \beta_{k,t} = 1 \\ \tilde{c}_{k,t}^{LNG}\bar{t}_r & \text{for } \alpha_{k,t} = 1 \end{cases} \tag{EC.1}$$

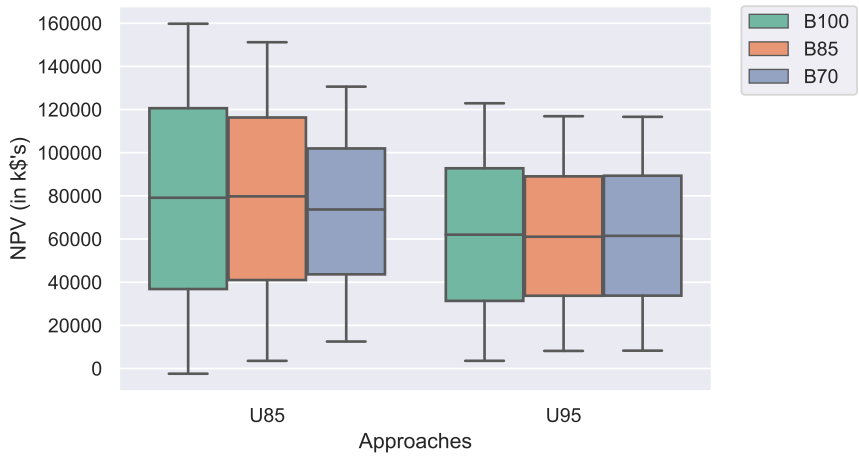


Figure EC.5 Boxplot of NPV for the instance V/12-Corr20

Prior to any installation of scrubbers or LNG retrofitting, that is, $\beta_{k,t} = \alpha_{k,t} = 0$, vessel k has to opt for VLSFO outside ECA for $\bar{t}_r - \bar{t}_r^{ECA}$ days at a fuel cost of $\tilde{c}_{k,t}^{VLSFO}(\bar{t}_r - \bar{t}_r^{ECA})$ per route r and switch to MGO within ECA for t_r^{ECA} days at a fuel cost of $\tilde{c}_{k,t}^{MGO}\bar{t}_r^{ECA}$ per route r. If a scrubber has been invested for vessel k in period t, that is, $\beta_{k,t} = 1$, then her fuel costs in route r, depending on the uncertain HSFO prices in that specific period t, amount to $\tilde{c}_{k,t}^{HSFO}\bar{t}_r$. If vessel k has retrofitted to use LNG in period t, that is, $\alpha_{k,t} = 1$, then the bunker fuel cost per route r becomes $\tilde{c}_{k,t}^{LNG}\bar{t}_r$.

Consequently, for any vessel k at a specific period t under $\alpha_{k,t}$ and $\beta_{k,t}$, the fuel cost in period t amounts to

$$\begin{aligned}
&\textstyle\sum_{r\in\mathcal{R}}\{[\tilde{c}_{k,t}^{VLSFO}(\bar{t}_r - \bar{t}_r^{ECA}) + \tilde{c}_{k,t}^{MGO}\bar{t}_r^{ECA}](1-\alpha_{k,t}-\beta_{k,t})x_{r,k,t} \\
&\qquad + \tilde{c}_{k,t}^{LNG}\bar{t}_r x_{r,k,t}\alpha_{k,t} + \tilde{c}_{k,t}^{HSFO}\bar{t}_r x_{r,k,t}\beta_{k,t}\}N_r \\
&= \textstyle\sum_{r\in\mathcal{R}}\{(\tilde{c}_{k,t}^{LNG} - \tilde{c}_{k,t}^{VLSFO})\bar{t}_r x_{r,k,t}\alpha_{k,t} \\
&\qquad + (\tilde{c}_{k,t}^{HSFO} - \tilde{c}_{k,t}^{VLSFO})\bar{t}_r x_{r,k,t}\beta_{k,t} \\
&\qquad + (\tilde{c}_{k,t}^{VLSFO} - \tilde{c}_{k,t}^{MGO})\bar{t}_r^{ECA} x_{r,k,t}(\alpha_{k,t}+\beta_{k,t}) \\
&\qquad + [\tilde{c}_{k,t}^{VLSFO}(\bar{t}_r - \bar{t}_r^{ECA}) + \tilde{c}_{k,t}^{MGO}\bar{t}_r^{ECA}]x_{r,k,t}\}N_r \\
&= \textstyle\sum_{r\in\mathcal{R}}\{-\tilde{\delta}_{k,t}^{VL}\bar{t}_r x_{r,k,t}\alpha_{k,t} - \tilde{\delta}_{k,t}^{VH}\bar{t}_r x_{r,k,t}\beta_{k,t} - \tilde{\delta}_{k,t}^{MV}\bar{t}_r^{ECA} x_{r,k,t}(\alpha_{k,t}+\beta_{k,t}) \\
&\qquad + [\tilde{c}_{k,t}^{VLSFO}(\bar{t}_r - \bar{t}_r^{ECA}) + \tilde{c}_{k,t}^{MGO}\bar{t}_r^{ECA}]x_{r,k,t}\}N_r
\end{aligned} \tag{EC.2}$$

where N_r is the number of loops covered per period per vessel under the existing weekly service of route r, calculated by $\lfloor \frac{\Delta t}{\bar{t}_r} \rfloor$.

Hence, the fleet's fuel cost saved in period t due to any technology investment can be expressed as

$$\begin{aligned}
CS_t(\tilde{\boldsymbol{\delta}}) = &\textstyle\sum_{k\in\mathcal{K}}\sum_{r\in\mathcal{R}}\{\tilde{\delta}_{k,t}^{VL}\bar{t}_r x_{r,k,t}(\alpha_{k,t}-\alpha_{k,0}) + \tilde{\delta}_{k,t}^{VH}\bar{t}_r x_{r,k,t}(\beta_{k,t}-\beta_{k,0}) \\
&+ \tilde{\delta}_{k,t}^{MV}\bar{t}_r^{ECA} x_{r,k,t}(\alpha_{k,t}+\beta_{k,t}-\alpha_{k,0}-\beta_{k,0})\}N_r.
\end{aligned}$$

EC.2.2 Linear reformulations of Equations (10) and (13c)

To handle the bi-linear terms in Equations (10) and (13c), we represent $w_{r,k,t} = \beta_{k,t} x_{r,k,t} \in \mathbb{B}$ and $u_{r,k,t} = \alpha_{k,t} x_{r,k,t} \in \mathbb{B}$ for any (r,k,t) triplet by standard linearization techniques as follows:

$$\begin{aligned} x_{r,k,t} - w_{r,k,t} \le 1 - \alpha_{k,t}, w_{r,k,t} \le x_{r,k,t}, w_{r,k,t} \le \alpha_{k,t} \\ x_{r,k,t} - u_{r,k,t} \le 1 - \beta_{k,t}, u_{r,k,t} \le x_{r,k,t}, u_{r,k,t} \le \beta_{k,t} \end{aligned}. \tag{EC.3}$$

EC.2.3 Linear reformulations of Equations (14) and (15)

We reformulate Equation (14) into linear constraints by introducing auxiliary decision variables $a_{r,k,t} \in \mathbb{B}$ and $b_{r,k,t} \in \mathbb{B}$ such that

$$\begin{aligned} & x_{r,k,t} \le a_{r,k,t} + b_{r,k,t} && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \alpha_{k,s} \le \tau_k^{LNG} + M_1 a_{r,k,t} && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \alpha_{k,s} \ge \tau_k^{LNG} - M_2(1 - a_{r,k,t}) + \epsilon && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \alpha_{k,s} \ge 1 - M_3 b_{r,k,t} && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \alpha_{k,s} \le 1 + M_4(1 - b_{r,k,t}) - \epsilon && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \end{aligned} \tag{EC.4}$$

where ϵ is a small positive number strictly less than 1 such as $\epsilon = 0.9$. Equation (EC.4) applies the big-M method again to ensure that both $a_{r,k,t}$ and $b_{r,k,t}$ equal 0 if $\sum_{s=1}^{t} \alpha_{k,s} \in [1, \tau_k^{LNG}]$ and $x_{r,k,t}$ is enforced to be 0; if $a_{r,k,t}$ or $b_{r,k,t}$ equals 1, $x_{r,k,t}$ is unaffected. In addition, it is sufficient to confine the selection of big M's in Equation (EC.4) to $M_1 \ge T$, $M_2 \ge \tau_k^{LNG} + 1$, $M_4 \ge T$, and $M_3 \ge 1$.

Similarly, Equation (15) can be represented by linear constraints as follows

$$\begin{aligned} & x_{r,k,t} \le (a'_{r,k,t} + b'_{r,k,t}) && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \beta_{k,s} \le \tau_k^{EGCS} + M_1 a_{r,k,t} && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \beta_{k,s} \ge \tau_k^{EGCS} - M_2(1 - a'_{r,k,t}) + \epsilon && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \beta_{k,s} \ge 1 - M_3 b'_{r,k,t} && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \\ & \textstyle\sum_{s=1}^{t} \beta_{k,s} \le 1 + M_4(1 - b'_{r,k,t}) - \epsilon && \forall k \in \mathcal{K}, t \in \mathcal{T}, r \in \mathcal{R} \end{aligned} \tag{EC.5}$$

where $a'_{r,k,t}, b'_{r,k,t} \in \mathbb{B}$ are the auxiliary decision variables introduced to apply the big-M method.

EC.2.4 Proof of Proposition 1

Proof of Proposition 1. Clearly, Equation (16) holds if $\beta_{k,t-1} = 1$ due to constraint (6). Otherwise, when $\beta_{k,t-1} = 0$, for any $t = 1, ..., T$, if $t > T - \underline{PP}_{k,t}^{EGCS}$, then

$$FC_k^{EGCS} \le \sum_{s=t+\tau_k^{EGCS}}^{T} \left(\delta_{k,s}^{VH}(\tilde{\boldsymbol{\nu}})\, \bar{t}_r + \delta_{k,s}^{MV}(\tilde{\boldsymbol{\nu}})\, \bar{t}_r^{ECA}\right) N_r, \forall r \in \mathcal{R}, \tilde{\boldsymbol{\nu}} \sim \mathbb{P} \in \mathcal{F}$$

by the definition of payback time in Model (5). Hence, $\beta_{k,t} = 1$ is strongly dominated by $\beta_{k,t} = 0$ due to the reduced NPV. In addition, Equations (13c) and (EC.5) implies that $\beta_{k,t} = 1$ is weakly dominated by $\beta_{k,t} = 0$ in terms of CUR. Similarly, we can show Equation (17). □

EC.2.5 Affine recourse approximations of Equation (18)

By affine recourse adaptation (19), we can rewrite constraints (18) for each $t \in \mathcal{T}$:

$$\sum_{(o,d)\in\mathcal{W}} g^{(o,d)} \left(\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) - \lambda \cdot Q_t^{(o,d)} \right) \geq 0 \qquad \forall \boldsymbol{Q}_t \in \mathcal{Q}_t$$

$$\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) - Q_t^{(o,d)} \leq 0 \qquad \forall \boldsymbol{Q}_t \in \mathcal{Q}_t, (o,d) \in \mathcal{W}$$

$$\sum_{(o,d)\in\mathcal{W}} \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \rho_{r,l}^{h^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) \leq \sum_{k\in\mathcal{K}} (V_k - \eta_k^{LNG} \alpha_{k,t} - \eta_k^{EGCS} \beta_{k,t}) x_{r,k,t} N_r$$

$$\forall \boldsymbol{Q}_t \in \mathcal{Q}_t, r \in \mathcal{R}, l = 1, ..., m_r,$$

which are equivalent to

$$\begin{aligned}
&\min_{\boldsymbol{Q}_t\in\mathcal{Q}_t} \sum_{(o,d)\in\mathcal{W}} g^{(o,d)} \left(\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) - \lambda \cdot Q_t^{(o,d)} \right) \geq 0 \\
&\max_{\boldsymbol{Q}_t\in\mathcal{Q}_t} \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) - Q_t^{(o,d)} \leq 0 \quad \forall (o,d) \in \mathcal{W} \\
&\max_{\boldsymbol{Q}_t\in\mathcal{Q}_t} \sum_{(o,d)\in\mathcal{W}} \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \rho_{r,l}^{h^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} Q_t^{(o,d)} \right) \leq \sum_{k\in\mathcal{K}} (V_k - \eta_k^{LNG} \alpha_{k,t} - \eta_k^{EGCS} \beta_{k,t}) x_{r,k,t} N_r \\
&\qquad \forall r \in \mathcal{R}, l = 1, ..., m_r.
\end{aligned} \tag{EC.6}$$

Then the above robust counterparts can be reformulated as linear constraints by taking the dual of the LHS of Equation (EC.6). For example, the first constraint in Equation (EC.6) is equivalent to

$$\begin{aligned}
&\sum_{(o,d)\in\mathcal{W}} g^{(o,d)} \left(\sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} \left(z_{t,0}^{h^{(o,d)}} + z_{t,1}^{h^{(o,d)}} \hat{Q}^{(o,d)} \right) - \lambda \cdot \hat{Q}^{(o,d)} \right) \geq \psi_t \Gamma_t + \sum_{(o,d)\in\mathcal{W}} (\xi_t^{(o,d)} \overline{\Delta Q}^{(o,d)} - \zeta_t^{(o,d)} \underline{\Delta Q}^{(o,d)}) \\
&-\psi_t \leq g^{(o,d)} \left(\lambda - \sum_{h^{(o,d)}\in\mathcal{H}^{(o,d)}} z_{t,1}^{h^{(o,d)}} \right) - \xi_t^{(o,d)} + \zeta_t^{(o,d)} \leq \psi_t \quad \forall (o,d) \in \mathcal{W} \\
&\xi_t^{(o,d)} \geq 0, \zeta_t^{(o,d)} \geq 0, \psi_t \geq 0 \quad \forall (o,d) \in \mathcal{W}
\end{aligned} \tag{EC.7}$$

where ψ_t, $\xi_t^{(o,d)}$, $\zeta_t^{(o,d)}$ are the dual variables. All the constraints in Equation (EC.6) can be reformulated similarly and we omit them for brevity.

EC.2.6 Proof of Proposition 2

Proof of Proposition 2. The first constraint in ARnO model can be written as

$$\sup_{\mathbb{Q}:\mathbb{Q}\ll\mathbb{P}} \mathbb{E}_{\mathbb{Q}} \left[B - NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}})) \right] - \mu D_{KL}(\mathbb{Q}||\mathbb{P}) \leq 0, \quad \forall \mathbb{P} \in \mathcal{F}$$

By the Donsker-Varadhan Variational Formula (see Dupuis and Ellis 2011, pp. 405), the above equation is equivalent to

$$\mu \ln \mathbb{E}_{\mathbb{P}} \left[\frac{B - NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}}))}{\mu} \right] \leq 0, \quad \forall \mathbb{P} \in \mathcal{F},$$

which can be further reformulated as $C_\mu[NPV(\boldsymbol{\delta}(\tilde{\boldsymbol{\nu}})) - B] \geq 0,\ \forall \mathbb{P} \in \mathcal{F}$, i.e., Equation (20). □

EC.2.7 Proof of Proposition 3

Proof of Proposition 3. We first note $NPV(\tilde{\boldsymbol{\delta}}) = \sum_{t=1}^{T} \gamma^t [CS_t(\tilde{\boldsymbol{\delta}}) - FC_t]$, then

$$\begin{aligned}
&C_\mu \left[NPV(\tilde{\boldsymbol{\delta}}) - B \right] \\
=&C_\mu \left[\sum_{t=1}^{T} \gamma^t [CS_t(\tilde{\boldsymbol{\delta}}) - FC_t] - B \right] \\
=&C_\mu \left[\sum_{t=1}^{T} \gamma^t CS_t(\tilde{\boldsymbol{\delta}}) \right] - B - \sum_{t=1}^{T} \gamma^t FC_t \\
=&C_\mu \left[\sum_{t=1}^{T} \gamma^t \sum_{k \in \mathcal{K}} \sum_{r \in \mathcal{R}} \tilde{\delta}_{k,t}^{VH} \bar{t}_r u_{r,k,t} + \tilde{\delta}_{k,t}^{VL} \bar{t}_r w_{r,k,t} + \tilde{\delta}_{k,t}^{MV} \bar{t}_r^{ECA} (w_{r,k,t} + u_{r,k,t}) \right] - B - \sum_{t=1}^{T} \gamma^t FC_t \\
=&C_\mu \left[\sum_{t=1}^{T} \gamma^t \sum_{k \in \mathcal{K}} \sum_{r \in \mathcal{R}} \sum_{\ell \in [3]} \tilde{\nu}_t^\ell \sum_{k \in \mathcal{K}} \sum_{r \in \mathcal{R}} \omega_k \left(\phi_{2\ell} \bar{t}_r u_{r,k,t} + \phi_{1\ell} \bar{t}_r w_{r,k,t} + \phi_{3\ell} \bar{t}_r^{ECA} (w_{r,k,t} + u_{r,k,t}) \right) \right] - B - \sum_{t=1}^{T} \gamma^t FC_t \\
=&\sum_{t=1}^{T} \gamma^t \sum_{\ell \in [3]} C_\mu \left[\tilde{\nu}_t^\ell \sum_{k \in \mathcal{K}} \sum_{r \in \mathcal{R}} \omega_k \left(\phi_{2\ell} \bar{t}_r u_{r,k,t} + \phi_{1\ell} \bar{t}_r w_{r,k,t} + \phi_{3\ell} \bar{t}_r^{ECA} (w_{r,k,t} + u_{r,k,t}) \right) \right] - B - \sum_{t=1}^{T} \gamma^t FC_t
\end{aligned}$$

where the last equality is from independent components of $\tilde{\boldsymbol{\nu}}$ since $C_\mu[\tilde{v}_1 + \tilde{v}_2] = C_\mu[\tilde{v}_1] + C_\mu[\tilde{v}_2]$ if $\tilde{v}_1$ and $\tilde{v}_2$ are independent. We introduce variables f_t^ℓ by Equation (21f) for convenience. Then the Equation (20) can be written as

$$\sum_{t=1}^{T} \gamma^t \sum_{\ell \in [3]} \inf_{\mathbb{P} \in \mathcal{F}} C_\mu \left[-\tilde{\nu}_t^\ell f_t^\ell \right] \geq B + \sum_{t=1}^{T} \gamma^t FC_t, \tag{EC.8}$$

We next focus on calculation of $\inf_{\mathbb{P}\in\mathcal{F}} C_\mu[-\tilde{\nu}_t^\ell f_t^\ell]$. Note that

$$\begin{aligned}
\sup_{\mathbb{P}\in\mathcal{F}} \mathbb{E}_{\mathbb{P}}\left[e^{\tilde{\nu}_t^\ell f_t^\ell/\mu}\right] \leq \quad & \inf_{\gamma\geq 0,\alpha,\beta} \quad \alpha+\beta\hat{\nu}_t^\ell+\gamma d_t^\ell \\
& \text{s.t.} \quad \alpha+\beta\nu_t^\ell+\gamma|\nu_t^\ell-\hat{\nu}_t^\ell| \geq e^{\nu_t^\ell f_t^\ell/\mu} \quad \forall \nu_t^\ell\in[\underline{\nu}_t^\ell,\overline{\nu}_t^\ell] \\
= \quad & \inf_{\gamma\geq 0,\alpha,\beta} \quad \alpha+\beta\hat{\nu}_t^\ell+\gamma d_t^\ell \\
& \text{s.t.} \quad \alpha+\beta\nu_t^\ell+\gamma(\nu_t^\ell-\hat{\nu}_t^\ell) \geq e^{\nu_t^\ell f_t^\ell/\mu} \quad \forall \nu_t^\ell\in[\hat{\nu}_t^\ell,\overline{\nu}_t^\ell] \\
& \qquad \alpha+\beta\nu_t^\ell+\gamma(\hat{\nu}_t^\ell-\nu_t^\ell) \geq e^{\nu_t^\ell f_t^\ell/\mu} \quad \forall \nu_t^\ell\in[\underline{\nu}_t^\ell,\hat{\nu}_t^\ell] \\
= \quad & \inf_{\gamma\geq 0,\alpha,\beta} \quad \alpha+\beta\hat{\nu}_t^\ell+\gamma d_t^\ell \\
& \text{s.t.} \quad \alpha+\beta\overline{\nu}_t^\ell+\gamma(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell) \geq e^{\overline{\nu}_t^\ell f_t^\ell/\mu} \\
& \qquad \alpha+\beta\hat{\nu}_t^\ell+\gamma(\hat{\nu}_t^\ell-\hat{\nu}_t^\ell) \geq e^{\hat{\nu}_t^\ell f_t^\ell/\mu} \\
& \qquad \alpha+\beta\underline{\nu}_t^\ell+\gamma(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell) \geq e^{\underline{\nu}_t^\ell f_t^\ell/\mu} \\
= \quad & \sup_{p_1,p_2,p_3\geq 0} \quad p_1 e^{\underline{\nu}_t^\ell f_t^\ell/\mu}+p_2 e^{\hat{\nu}_t^\ell f_t^\ell/\mu}+p_3 e^{\overline{\nu}_t^\ell f_t^\ell/\mu} \\
& \text{s.t.} \quad p_1+p_2+p_3=1 \\
& \qquad p_1\underline{\nu}_t^\ell+p_2\hat{\nu}_t^\ell+p_3\overline{\nu}_t^\ell=\hat{\nu}_t^\ell \\
& \qquad p_1(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)+p_3(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell) \leq d_t^\ell
\end{aligned}$$

where the first inequality is by weak duality, the second equality is because the optimal solution of a convex maximization problem is attained at the boundary, and the third equality is due to linear optimization strong duality. Clearly, the worst-case distribution is attained by a three-point distribution with probability mass $\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}$, $\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}$, and $1-\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}-\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}$ on $\underline{\nu}_t^\ell$, $\overline{\nu}_t^\ell$, $\hat{\nu}_t^\ell$ by solving the last linear optimization problem in the above bound. Therefore,

$$\sup_{\mathbb{P}\in\mathcal{F}} \mathbb{E}_{\mathbb{P}}\left[e^{\tilde{\nu}_t^\ell f_t^\ell/\mu}\right] = \frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}\exp\left(\frac{\underline{\nu}_t^\ell f_t^\ell}{\mu}\right)+\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\exp\left(\frac{\overline{\nu}_t^\ell f_t^\ell}{\mu}\right)+\left(1-\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}-\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\right)\exp\left(\frac{\hat{\nu}_t^\ell f_t^\ell}{\mu}\right).$$

We introduce auxiliary variables ξ_t^ℓ such that $-\xi_t^\ell \leq \inf_{\mathbb{P}\in\mathcal{F}} C_\mu[-\tilde{\nu}_t^\ell f_t^\ell]$, then the term $\inf_{\mathbb{P}\in\mathcal{F}} C_\mu[-\tilde{\nu}_t^\ell f_t^\ell]$ in Equation (EC.8) can be replaced by $-\xi_t^\ell$ with an additional constraint

$$\mu\ln\left(\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}\exp\left(\frac{\underline{\nu}_t^\ell f_t^\ell}{\mu}\right)+\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\exp\left(\frac{\overline{\nu}_t^\ell f_t^\ell}{\mu}\right)+\left(1-\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}-\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\right)\exp\left(\frac{\hat{\nu}_t^\ell f_t^\ell}{\mu}\right)\right) \leq \xi_t^\ell,$$

By introducing auxiliary variables $\boldsymbol{\theta}_t^\ell$ we can reformulate the above constraint as

$$\begin{aligned}
& \frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}\theta_{t,1}^\ell+\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\theta_{t,2}^\ell+\left(1-\frac{d_t^\ell}{2(\hat{\nu}_t^\ell-\underline{\nu}_t^\ell)}-\frac{d_t^\ell}{2(\overline{\nu}_t^\ell-\hat{\nu}_t^\ell)}\right)\theta_{t,3}^\ell \leq \mu \\
& \mu\exp\left(\frac{\underline{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right) \leq \theta_{t,1}^\ell \\
& \mu\exp\left(\frac{\overline{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right) \leq \theta_{t,2}^\ell \\
& \mu\exp\left(\frac{\hat{\nu}_t^\ell f_t^\ell-\xi_t^\ell}{\mu}\right) \leq \theta_{t,3}^\ell,
\end{aligned}$$

i.e., Equations (21b)-(21e). Combining them with constraints (21f), we conclude the proof.

EC.2.8 Proof of Proposition 4

Proof of Proposition 4 Note that $(x_1, x_2, x_3) \in \mathcal{C}_{\exp}$ if and only if $(x_2, x_1, x_3) \in \mathcal{C}_{re}$ where $\mathcal{C}_{re}$ is a relative entropy cone defined by

$$\mathcal{C}_{re} = \mathrm{cl}\left\{(y_1, y_2, y_3) : y_1 \ln(y_1/y_2) \leq y_3, y_1 > 0, y_2 > 0\right\}.$$

The rest of the proof follows from Theorem 2 in Fawzi et al. (2019).

EC.3 Data Calibration

EC.3.1 Ambiguity set construction

Data source. The fuel price data is not publicly available and ocean carrier subscribes specialized platforms to gain access to bunker indices. The fuel price data used in this study was obtained from a UK based energy and commodities information platform, S&P Global Platts[12] from Jan 1, 2020 to May 32, 2021 (including 369 trading days) based on the free on board (FOB) price delivered to Singapore (unit: USD per MT). We retrieve the Henry Hub Natural Gas Spot Price (unit: USD per mmBtu) from the EIA database for the same period[13].

Data processing. To unify the units for different types of energy. We adopt the industry practical assumptions as follows[14].

- Gas lower heating value (LHV) has to be assumed for ship fuel. The Henry Hub Natural Gas Spot Price indicates upper heating value (UHV) which gives about 10% lower LNG prices than indicated here for LHV prices.
- Liquefaction costs have to be added to Henry Hub Natural Gas Spot Price. We adopt a cost of 3 USD per mmBtu.
- The fuel conversion rate for ship fuel as: 1 MT $\approx$ 38.4 mmBtu.

We calculate the historical price differences $\Delta c_t^{VL}, \Delta c_t^{VH}$ and Δc_t^{MV} accordingly.

The ambiguity sets. To eliminate the strong correlations among the three price differences, we conducted principal component analysis (PCA) to pick out orthogonal components, denoted as ν_t^1, ν_t^2 and ν_t^3 with the coefficient indicated below.

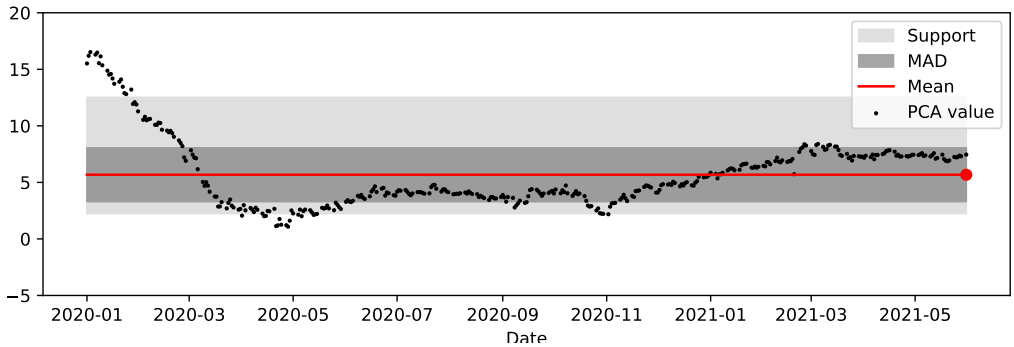
$$\begin{bmatrix} \Delta c_t^{VL} \\ \Delta c_t^{VH} \\ \Delta c_t^{MV} \end{bmatrix} = \begin{bmatrix} 0.8724 & -0.4400 & -0.2127 \\ 0.4882 & 0.7654 & 0.4193 \\ -0.0217 & -0.4696 & 0.8826 \end{bmatrix} \begin{bmatrix} \nu_t^1 \\ \nu_t^2 \\ \nu_t^3 \end{bmatrix}$$

We next construct ambiguity sets for the three components according to their mean, support and MAD, as illustrated in Figure EC.6 and summarized in Table EC.1

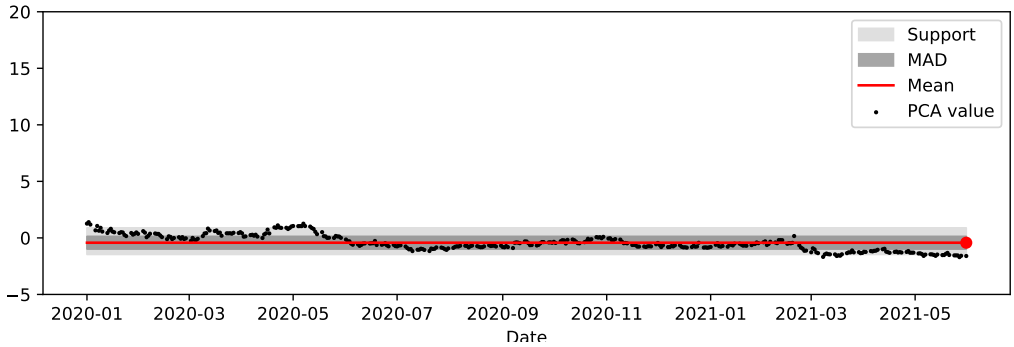
[12] See `https://www.spglobal.com/platts/en`

[13] Source: `https://www.eia.gov/dnav/ng/hist/rngwhhdm.htm`; accessed Jan 9, 2021

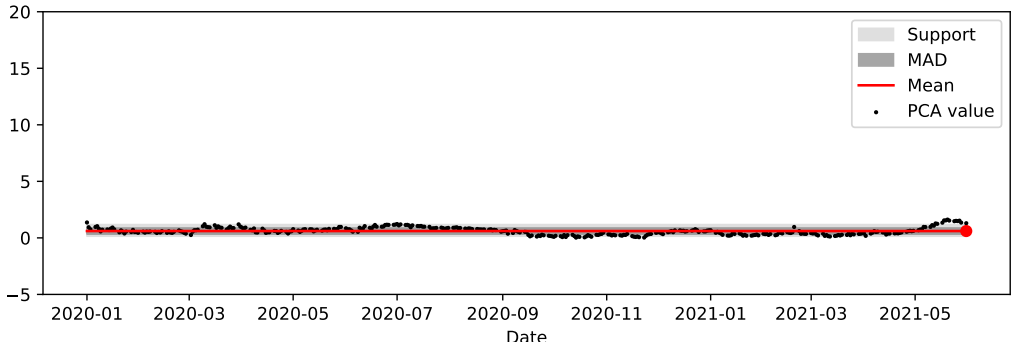
[14] Source: `https://www.dnv.com/maritime/insights/topics/lng-as-marine-fuel/current-price-development-oil-and-gas.html`; accessed Jan 9, 2021



(a) ν_t^1



(b) ν_t^2



(c) ν_t^3

Figure EC.6 Ambiguity sets (mean, support and MAD) of the three components

a. (Mean) We approximate the mean by the historical mean of each component (see Figure EC.6)[15].

b. (Support) We construct a support set that covers 90% of the observations. In particular, we net off the mean from a PCA component to get the residual. We then obtain the 5%- and 95%-quantile of the residual.

[15] We assume a time-independent mean. To verify this, we identify the trend as a time-dependent function of date t through a linear regression and found that coefficients associated with t are -0.0013, -0.0052 and -0.0006 for ν_t^1, ν_t^2, ν_t^3, respectively (p-value = 0.000).

c. (MAD) We construct the MAD by calculating the mean of the residual explained above.

Table EC.1 Summary statistics for the three components

	Mean	Support[a.]	MAD
$\tilde{\nu}_t^1$	5.918	$-3.428/+6.844$	2.373
$\tilde{\nu}_t^2$	0.529	$-1.034/+1.313$	0.559
$\tilde{\nu}_t^3$	0.725	$-0.453/+0.566$	0.263

[a.] For any component whose mean is x, we denote by $-y/+z$ the support $[x-y, x+z]$. For example, ν_t^1 has support $[5.918-3.428, 5.918+6.844]$.

EC.3.2 Liner shipping service

The Pacific Northwest service consists of four shipping routes, NP1, NP2, NP3 and NP4, as illustrated in Figure EC.3. We consider the set of four routes as $\mathcal{R}$. The detailed service information about each route, including frequency, number of vessels employed on each route (or fleet size), ports of call, and the transit time between any two ports are retrieved from the APL official website (Source: `http://www.apl.com/our-offerings/services`, accessed July 2020), as shown in the following table.

Table EC.2 Routes information for APL Asia-North America West Coast service

Route	NP1[a]		NP2[a]		NP3[a]		NP4[a]	
Frequency	Weekly		Weekly		Weekly		Weekly	
Fleet size	6 vessels		6 vessels		6 vessels		6 vessels	
Ports of call	Ports	TransitTime	Ports	TransitTime	Ports	TransitTime	Ports	TransitTime
	YANTIAN	0	HONG KONG	0	QINGDAO	0	SHEKOU	0
	XIAMEN	1	YANTIAN	1	SHANGHAI	1	HONG KONG	1
	NINGBO	2	NINGBO	3	NINGBO	3	YANTIAN	1
	SHANGHAI	3	SHANGHAI	5	KAOHSHIUNG	5	KAOHSHIUNG	3
	BUSAN	6	PRINCE RUPERT	17	YANTIAN	7	VANCOUVER	16
	SEATTLE	19	VANCOUVER	21	TACOMA	22	SEATTLE	19
	VANCOUVER	21	HONG KONG	40	VANCOUVER	25	BUSAN	36
	GWANGYANG	38			TOKYO	36	KAOHSHIUNG	40
	YANTIAN	41			OSAKA	38	SHEKOU	41
					QINGDAO	41		
Num of ports	8		6		9		8	
Duration	42 days		42 days		42 days		42 days	
Distance[b]	12023 nm		11927 nm		12887 nm		11973 nm	
ECA dist[b]	2962 nm		3104 nm		2637 nm		1699 nm	
Time at sea ($\bar{t}_r$)[b]	20 days and 21 hours		20 days and 17 hours		22 days and 9 hours		20 days and 19 hours	
ECA time ($\bar{t}_r^{ECA}$) [b]	5 days and 3 hours		5 days and 9 hours		4 days and 14 hours		2 days and 21 hours	

[a.] Source: `https://www.apl.com/products-services/line-services/flyer/CPNW` (NP1); `https://www.apl.com/products-services/line-services/flyer/TWPKS` (NP2) ; `http://www.apl.com/products-services/line-services/flyer/NWX` (NP3); `http://www.apl.com/products-services/line-services/flyer/NWX` (NP4) assessed July 2020.
[b.] All distance related information, including "Distance" and "ECA distance" are obtained from `https://app.aquaplot.com/app/route`. The corresponding time, including the "time at sea" and "ECA time" are calculated based on an average speed of 24 knot.

EC.3.3 Fleet information

Twenty-four heterogeneous vessels are employed on the four routes, six vessels per route, in order to maintain the current weekly service provided that the total duration to finish each route amounts to six weeks. We consider these 24 vessels as our fleet $\mathcal{K}$. The information on each vessel can be found from the same website, including vessel capacity in TEU and the current route a particular vessel is employed to, as shown in Table EC.3.

Table EC.3 Fleet information and payback time

Index	Capacity (in TEU)	Current route assigned	Payback time (in years)[a]		$\underline{PP}_k$ (in years)[b]	
			EGCS	LNG	EGCS	LNG
1	5652	NP3	3.75	16.5	2	7.75
2	5652	NP3	3.75	16.5	2	7.75
3	5652	NP3	3.75	16.5	2	7.75
4	5652	NP3	3.75	16.5	2	7.75
5	5750	NP4	4	17.75	2	8.25
6	5828	NP4	4	17.75	2	8.25
7	5888	NP3	3.75	16.5	2	7.75
8	5888	NP4	4	17.75	2	8.25
9	5888	NP4	4	17.75	2	8.25
10	5888	NP4	4	17.75	2	8.25
11	7024	NP4	4	17.75	2	8.25
12	8102	NP3	2.75	12.5	1.5	6
13	8586	NP1	2.5	12.75	1.5	6.25
14	8586	NP1	2.5	12.75	1.5	6.25
15	8586	NP1	2.5	12.75	1.5	6.25
16	8888	NP1	2.5	12.75	1.5	6.25
17	8888	NP2	2.75	13	1.5	6.25
18	8888	NP2	2.75	13	1.5	6.25
19	8888	NP2	2.75	13	1.5	6.25
20	9200	NP2	2.5	12.25	1.5	6
21	9572	NP2	2.5	12.25	1.5	6
22	9956	NP1	2.5	12.25	1.5	6
23	10036	NP2	2.5	12.25	1.5	6
24	10106	NP1	2.5	12.25	1.5	6

[a.] Calculated from Model (5) under expected fuel prices.
[b.] Greatest lower bound of payback time, defined and calculated according to Proposition 1.

EC.3.4 The service network

Apart from the liner shipping service information on $\mathcal{R}$ and $\mathcal{P}$, as indicated in Table EC.2, we set the parameter value of $\mathcal{W}$ and $\mathcal{H}^{(o,d)}$ according to the following procedure. First, we identify all the (o, d) pairs and the corresponding path(s) that can be fulfilled by any route without any transshipment. Next, we observe that YANTIAN is covered by all four routes, so we consider YANTIAN as a transshipment hub and identify all the (o, d) pairs that can be fulfilled by any two routes jointly through a transshipment at YANTIAN. Finally, to reflect reality, we delete all the paths that go across the pacific ocean more than once. For example, the voyage from SHANGHAI to VANCOUVER can be fulfilled by (i) NP1 directly (ii) NP2 directly (iii) NP3 directly (iv) NP3 and NP1 jointly with a transshipment at YANTIAN (v) NP3 and NP2 jointly with a transshipment

at YANTIAN (vi) NP3 and NP4 jointly with a transshipment at YANTIAN. We simulate the value of demand on any (o, d) pair according to leg demand that is randomly generated from the interval 0.1 to 1.0 times the capacity of the largest ship employed on the routes involved. Similar demand generating procedures are adopted in Agarwal and Ergun (2008) and such proportions are also used in Fagerholt (1999). Demand sizes on each (o, d) pair, namely $Q^{(o,d)}$, are assumed through a network flow problem that maximizes the total number of containers carried subject to the constraints of leg flows (approximated by leg demands). We consider this set of demand, denoted by $\{\hat{Q}^{(o,d)}\}_{(o,d)\in\mathcal{W}}$, as a base case and scale it by different factors to reflect the variations of demand over time.

EC.3.5 Scrubber installation and LNG retrofitting parameters

We set the value of IMO 2020 related parameters as shown in Table EC.4. Let $\eta_k^{EGCS} = \lceil 0.4\% \times V_k \rceil$ and $\eta_k^{LNG} = \lceil 3\% \times V_k \rceil$, corresponding to the loss of TEU slots to install scrubbers and retrofit vessels to use LNG, respectively. We also assume a lead time of $\tau_k^{LNG} = 5$ for LNG retrofitting and $\tau_k^{EGCS} = 2$ for scrubber installation in any vessel k, corresponding to 5 and 2 quarters, respectively.

Table EC.4 Fixed cost for scrubber installation and LNG retrofitting

Vessel Size (in TEU)	Vessel index k	FC_k^{EGC} [a.] (in m$)	FC_k^{LNG} [b.] (in m$)	Fuel consumption ω_k [c.] (in '000 mmBtu/day)
5000∼6000	13, 14, 16∼24	3.8	25	3.8
7000∼8000	15	3.9	28	5.8
8000∼9000	1, 2, 5, 6, 8, 10, 11	3.95	30	6.5
9000∼10200	2, 3, 7, 9, 12	4	32	7.3

[a.] EGCS installation fixed costs are estimated by taking the Very Large Crude Carrier (VLCC) as a reference. Note that, for VLCCs (160,000–319,999 DWT), the cost of fitting an open loop scrubber in a newbuild ship is around $2.5-$3.0 million, whereas the cost of retrofitting a scrubber on an existing VLCC will be $4-$4.5 million. Source: `https://www.drewry.co.uk/maritime-research-opinion-browser/maritime-research-opinions/the-dilemma-of-fitting-scrubbers`.

[b.] Estimated according to newsletters, see `https://www.pacificgreen-group.com/articles/why-marine-exhaust-scrubbers-are-cheaper-lng-conversion` and `https://shipandbunker.com/news/world/738570-hapag-lloyd-lng-bunkering-retrofit-for-boxship-to-cost-35-million`.

[c.] Fuel consumption data are estimated according to a survey study. Source: `https://transportgeography.org/contents/chapter4/transportation-and-energy/fuel-consumption-containerships/`; accessed Jan 9, 2021